

Customer Segmentation for Personalised Market Basket Analysis: An RFM-based K-Means and Apriori Algorithm in E-Commerce Retail

Vinicius Carrarini Cabral
Higher Education in Data Analytics
National College of Ireland
Dublin
x23429895@student.ncirl.ie

Abstract: One of the main challenges organizations face is extracting actionable knowledge from the vast volumes of data collected daily. Data mining techniques have emerged as an approach to assess and transform raw data into meaningful insights. This study applies two well-established data mining techniques in the e-commerce domain – K-Means clustering and association rule mining – implementing customer segmentation and market basket analysis as analytical applications to evaluate whether a segmented analytical pipeline produces more informative purchasing patterns than aggregate analysis alone. An RFM-based K-Means clustering algorithm was applied using the Online Retail II dataset to identify distinct customer segments based on Recency, Frequency, and Monetary purchasing patterns. Following this, Apriori and FP-Growth were applied to each resulting segment to extract frequent itemsets and association rules. Both algorithms produced identical results, with minimal runtime differences at this dataset's scale — Apriori was used for the subsequent analysis. The clustering algorithm identified four distinct segments, later refined to six through manual analytical refinement. Association rule mining across all six segments produced strong evaluation metrics, yet producing comparable results to the aggregated dataset baseline. The main findings demonstrated that the segmented applications revealed a greater volume of association rules, uncovering distinct product purchasing patterns within each customer segment. These findings suggest that the combined pipeline provides a more informative analysis of customer purchasing behaviour for e-commerce retailers.

Keywords—Customer Segmentation, Market Basket Analysis, K-Means Clustering, Association Rule Mining, Apriori Algorithm, FP-Growth, RFM Analysis, Data Mining, E-commerce, Unsupervised Learning, Strategic Marketing, Product Recommendation, Retail.

I. INTRODUCTION

A. Background and motivations

Nowadays one of the most valuable resources is knowledge acquired from data. We live in an era where there is a large amount of data being collected daily, and most businesses underutilize data for strategic purposes. E-commerce companies have been facing increasing competition in the last few years, making the use of techniques to extract and transform data into valuable information essential for growth. In this scenario, Data

Mining (DM) has emerged as a process to transform raw data into organized knowledge. It has become critical for Business Intelligence (BI) to acquire more information on their whole organization, such as customers, market, supply, resources, and competitors (Han, Kamber and Pei, 2012).

Market Basket Analysis (MBA) is a DM application that analyses frequently purchased items to explore associations and correlations between products in large transactional data. Likewise, Customer Segmentation Analysis (CSA), another DM application, is applied in BI to create subsets of customers based on shared characteristics (Han, Kamber and Pei, 2012). Both applications are widely used in e-commerce for cross-selling, product recommendations, product bundling, personalised marketing and customer retention, increasing overall sales and more.

This project aims to develop a Customer Segmentation followed by a Market Basket Analysis on each customer subset of an online retail transactional dataset, evaluating whether a personalised MBA approach produces more informative purchasing patterns than aggregate analysis alone.

B. Research Questions

- RQ1 - How can Recency, Frequency, and Monetary (RFM) based features and K-Means clustering algorithm be applied to identify the optimal number of customer segments in an e-commerce retail dataset, and what behavioural characteristics define each segment?
- RQ2 - What frequent itemsets and association rules can be discovered within each customer segment, and do they demonstrate meaningful patterns when evaluated through support, confidence, and lift?
- RQ3 - Does applying MBA at the customer segment level produce stronger and more distinct association rules compared to applying it on the full dataset?

C. Objectives

1. Data Preparation & Feature Engineering: To clean, prepare and engineer Recency, Frequency, and Monetary features to ensure suitability for clustering applications.
2. Customer Segmentation: To apply the K-Means clustering algorithm, identifying the optimal number of customer segments through standard evaluation metrics.

3. **Profiling:** To interpret each customer segment and assign labels based on Recency, Frequency, and Monetary patterns.

4. **Market Basket Analysis:** To apply association rules mining to each customer segment, extracting frequent itemsets that represent common product combinations purchased within each group.

5. **Rule Evaluation:** To assess the quality of association rules using support, confidence, and lift metrics, identifying those with practical significance.

6. **Association Rules Comparison:** To compare the association rules generated from the full dataset against those generated at the customer segment level, identifying whether meaningful differences and new patterns exist between them.

II. TECHNICAL BACKGROUND AND RELATED WORK

MBA is traditionally applied to entire customer populations, but purchasing patterns differ across demographics, spending level and other customer features. Lim (2021) explored the idea of applying customer segmentation to improve MBA, noting that there is limited research on the topic. This study addresses that gap by proposing a pipeline that combines customer segmentation with association rule mining to explore a personalised MBA for each customer segment.

1) Customer Segmentation

The term ‘market segmentation’ was first introduced by Smith (1956), who described it as an adjustment of market strategy by identifying smaller groups with similar characteristics within a diverse market. In the context of customer data analysis, this concept is known as customer segmentation — the process of dividing the customer base into groups that share similar behavioural or characteristic patterns. The purpose is to use customer pattern information to increase retention, cross-selling, upselling, and referral outcomes (Hughes, 2012).

Clustering is an unsupervised learning data mining technique defined as the process of partitioning data points into groups based on shared attributes, rather than predefined categories (Aggarwal and Reddy, 2014). Several clustering techniques have been proposed in the literature, each with distinct assumptions and trade-offs, including probabilistic, distance-based, and density-based methods (Aggarwal and Reddy, 2014). The clustering algorithm conducted in this study for customer segmentation is the distance-based K-Means clustering.

K-Means is a widely used clustering algorithm in data mining literature for its simplicity and ease of implementation. It is a partitional clustering that, through an iterative process, assigns data points to a predefined number of clusters (k) based on proximity to centroids (Aggarwal and Reddy, 2014). Alves Gomes and Meisen (2023) identified K-Means clustering as the most frequently used application for customer segmentation, approximately 39% of all papers the authors reviewed. K-Means has been shown to be very efficient, with minimal computational time and easy to use, but it has not always been the best application. As identified by Ling and Weiling (2025) variations of K-Means demonstrated better performance (K-Means++ and K-Medoids) in a comparative study to uncover advantages of using unsupervised clustering using different methods to

enhance customer segmentation. In addition, John, Shobayo and Ogunleye (2023) achieved good results using different models, including K-Means, aiming to develop a customer segmentation model for a retail market industry, but other models had better performance based on their evaluation process.

The selection of an appropriate clustering depends on both objectives and the characteristics of the dataset, and as Alves Gomes and Meisen (2023) identified in their review of customer segmentation methods for e-commerce targeting, no single algorithm consistently outperforms all others in every context. Although K-Means has been the most frequently applied method across all use cases, the authors acknowledge that it is not necessarily the best solution, and the algorithm selection must be guided by the specific data and segmentation goal.

Two main limitations of K-Means require some attention. First, the algorithm is sensitive to outliers and noisy data, which can degrade both clustering quality and predictive accuracy. Second, K-Means is sensitive to initial centroid placement, affecting stability on cluster assignments by random initialisation. The first limitation requires attention on the preprocessing step. Ling and Weiling (2025) applied IQR-based removal, which effectively reduced the influence of extreme values. However, this method can be aggressive when applied to retail customer databases, where high-value customers may be incorrectly removed. The second limitation was addressed by some authors through the application of K-Means++, an improved initialisation strategy – its benefit was confirmed by Ling and Weiling (2025), who found K-Means++ to be the best performing algorithm in their comparative work.

In order to address this weakness, a less aggressive method was used to remove only extreme values in highly skewed features, and K-Means++ was applied to check if the model would be improved. Wongoutong (2024) identified that when applying a K-Means clustering, a crucial step is feature scaling for a more accurate and reliable cluster assignment. *MinMaxScaler* and *RobustScaler* were examined to determine the most appropriate scaling approach.

An important preliminary step in K-Means application is the selection of features to be used for clustering model. Depending on the dataset, many features can be irrelevant or even limit the analysis, therefore filtering relevant features can improve learning quality, memory consumption, and computational time. Feature selection aims to overcome any high-dimensional dataset or noisy and redundant information, ensuring only relevant variables are selected (Aggarwal and Reddy, 2014).

RFM has been used as a marketing strategy for over 60 years to analyse customer behaviour and customer segmentation based on their purchase patterns (Hughes, 2012). Alves Gomes and Meisen (2023) identified that RFM-analysis is the most used feature selection method for customer representation since it can be applied to any kind of purchase data, it is simple to calculate and can handle any size of dataset without scalability issues. The authors highlighted that one of the weaknesses is that RFM does not consider the whole information of the customer, such as browsing, review, engagement, and demographic behaviour. Therefore, this study will adopt RFM analysis since it provides a simple,

practical and well-established approach on customer segmentation applications.

Clustering evaluation in unsupervised learning is challenging, as the lack of the ground truth makes it difficult to establish a consensus on the best validation metrics. Alves Gomes and Meisen (2023) did not find the best evaluation metric across their literature review. The authors classified the approaches found into several groups: some researchers analysed the clusters based on their context and plausibility; others used statistical significance tests to determine whether the data points are randomly distributed; An analysis of variance (ANOVA) was applied to check whether the values of variables were different from each other; Silhouette analysis evaluated both consistency within a cluster and the optimal number of clusters; Different indices were used to evaluate the cluster performance as Davies-Bouldin (DB), Calinski-Harabasz (CH), and Xie-Beni(XB) indexes; Additional approaches noted by the authors included error measures, information criteria, and others. The Elbow metric was also acknowledged as a common approach for determining the optimal number of clusters, commonly used in combination with other metrics.

Wasilewski (2024) identified the same problem of selecting an optimal method for evaluating clustering applications, reporting that a model with a multi-factor approach is more robust and reliable than a single-factor evaluation. Following the authors' recommendation, this study adopted a multi-factor evaluation framework: Silhouette score, one of the most popular and widely adopted validation metrics in clustering literature (Alves Gomes and Meisen, 2023), combined with Elbow method to support the selection of the optimal number of clusters; Davies-Bouldin index (DBI) - a commonly used index metric that assesses cluster compactness and separation, complementing the silhouette score in assessing cluster quality; R², conducted on JASP-stats, measures how well the model fits into the data, however does not indicate whether k inflation may be present; Finally, a cluster profile analysis was conducted to check each cluster results through its distribution balance across segments, visual exploration, and mean RFM values per cluster.

2) Market Basket Analysis

After establishing the customer segmentation framework, this section introduces Market Basket Analysis as the second phase of the proposed pipeline.

Market Basket Analysis is a data mining technique that analyses customer buying behaviour, providing businesses with actionable insights for the development of targeted marketing strategies based on item purchasing patterns (Han, Kamber and Pei, 2012). Alawadh and Barnawi (2022) identified six business applications relevant to this domain: predicting customer purchase behaviour, understanding customer demand, product positioning, designing target promotions, cross-selling strategies, and consumer segmentation.

Items frequently purchased together are represented in the form of association rules. Alawadh and Barnawi (2022) found in their paper review that association rules are the most suitable and widely used mining technique for MBA, particularly for retail and e-commerce transactional datasets. For this reason, to identify association rules within each customer segment, two of the most adopted mining algorithms were applied: Apriori, first proposed by Agrawal, Imieliński and Swami (1993) and FP-Growth, introduced by Han, Pei and Yin (2000).

Apriori is one of the most widely adopted algorithms for association rule mining (Alawadh and Barnawi, 2022), employing an iterative level-wise search to identify frequent itemsets and subsequently deriving association rules that meet a user-defined minimum support and confidence thresholds (Agrawal, Imieliński and Swami, 1993). FP-growth was developed to address two limitations of Apriori: the generation of a large number of candidate itemsets and the need to repeatedly scan the whole dataset. Instead of generating candidates, FP-Growth compresses the dataset into a frequent pattern tree and divides it into a set of conditional datasets, mining each of them separately without revisiting the full dataset (Han, Kamber and Pei, 2012).

Alawadh and Barnawi (2022) compared Apriori and FP-Growth across different studies and found that generally they

Table 1- Summary of related literature on customer segmentation.

PAPER	DATASET	FEATURE ENGINEERING	MODELS	EVALUATION METRICS	BEST RESULTS
Wu et al. (2022)	Kaggle Brazilian e-commerce	RFM + C and V	Improved K-Medoids, K-Means++, Spectral clustering, K-Medoids	Accuracy, Runtime, CH index, Elbow Method	Improved K-Medoids
Ling & Weiling (2025)	Kaggle Online Retail	RFM	K-Means++, K-Means, Mean Shift, GMM, K-Medoids	Silhouette, DBI, Elbow method, MAE/MSE/R ²	K-Means++
Wasilewski et al. (2024)	AIM2 platform clickstream	Behavioural features	K-Medians, BIRCH, K-Means, Agglomerative, Genetic	Silhouette, DBI, Elbow CH index, Cluster Spread index, Min cluster size	K-Medians
John et al. (2023)	UCI Online Retail (UK)	RFM	GMM + PCA, K-Means, BIRCH, Agglomerative, DBSCAN	Silhouette Score	GMM
Herman et al. (2022)	Financial statements	Financial ration features	K-Medoid, K-Means, Hierarchical/Ward's	Elbow Method, Dendogram, Intra-cluster size balance, Mean (ROS/ROA/ROE)	K-Medoids

Note: RFM = recency, frequency, monetary; C = add-to-cart count; V = favourites count; DBI = Davies-Bouldin Index; CH = Calinski-Harabasz; BIRCH = Balanced Iterative Reducing and Clustering using Hierarchies; GMM = Gaussian Mixture Model; PCA = Principal Component Analysis; MAE = Mean Absolute Error; MSE = Mean Squared Error; R² = Coefficient of Determination; ROA = Return on Assets; ROE = Return on Equity; ROS = Return on Sales

Table 2 - Summary of related literature on Market Basket Analysis

PAPER	DATASET	FEATURE ENGINEERING	MODELS	EVALUATION METRICS	BEST RESULTS
Hossain et al. (2019)	French Retail Store; Bakery Shop - Kaggle	Apriori; FP-Growth	min_support = 1%; min_confidence = 50%	Execution time, number of frequent itemsets, number of association rules	FP-Growth was faster than Apriori. Both presented identical association rules. Top 55% top-selling product reduction preserved equivalent rules with lower computation
Alawadh & Barnawi (2022)	Survey of 20 studies (2016-2020) across different industries	Apriori; FP-Growth; Genetic Algorithms; Neural Networks; Swarm Intelligence	Vary across studies	Support, Confidence, Lift, Leverage, Conviction, Interestingness, Comprehensibility	ARL are not limited to retail. MBA applications can be used in healthcare, sports, hospitality and more. Hybrid approaches outperform traditional algorithms on large, diverse datasets.
Patil et al. (2023)	Retail grocery dataset - Kaggle	Apriori; FP-Growth; ECLAT	min_support = 0.1-0.5; min_confidence = 50%	Execution time, Support, Confidence, Lift, Leverage, Conviction	FP-Growth was the fastest of the three. Apriori produced the most comprehensive and highest-quality association rules. ECLAT was notably slower.
Riyadi et al. (2019)	E-commerce transaction data - XYZ company, Indonesia	Apriori; FP-Growth	min_support = 2%; min_confidence = 90%	Lift Ratio; Support x Confidence Accuracy Score; Software Quality Assurance	Apriori produced higher accuracy rules.
Jirapatsil & Phumchusri (2022)	E-commerce warehouse - Thai fresh products company	Apriori	min_support = 10%; min_confidence applied post-generation	Support, Confidence, Lift, Average picking distance per order	Association Rule results applied to operational improvement reduced the average warehouse picking size by around 50% after product relocation.

Note: min = minimum; ECLAT = Equivalence Class Clustering and Bottom-up Lattice Traversal; FP-Growth = Frequent Pattern Growth

produce similar results, but FP-Growth outperforms Apriori in execution time on large databases. Supporting this finding, Dwiputra *et al.* (2023) applied both algorithms to an MBA for a pharmaceutical transactional dataset and found that both algorithms generated the same number of rules and identical metric values. Apriori demonstrated the lowest memory consumption, while FP-Growth had the faster execution time. The authors noted that algorithm selection depends on objectives, itemset size, data scale, and computational capabilities.

Support and confidence are two primary measures used to evaluate association rules. Support measures how frequently a rule occurs in the dataset, while confidence measures the reliability of the rule. In order to generate meaningful association rules, both measures must meet a minimum threshold, usually defined by users or domain experts (Han, Kamber and Pei, 2012). The threshold selection is one of the limitations Alawadh and Barnawi (2022) highlighted that setting a high threshold risks excluding valuable patterns, while a low threshold produces weak and irrelevant rules. Han, Kamber and Pei (2012) observed that association rules may still be misleading or uninteresting even when they satisfy the minimum support and confidence thresholds, highlighting the limitations of relying only on the support-confidence framework. To address this, the authors introduced lift as a correlation measure, which evaluates the correlation between the antecedent and consequent by comparing their observed co-occurrence against what would be expected under statistical independence. Alawadh and Barnawi (2022) similarly identified several additional measures, noting that deeper insights can be obtained through metrics such as lift, leverage, and conviction. Additionally, visualization tools such as grouped matrix visualisations are an effective way to present large sets of association rules, where antecedents are shown as columns, consequents as rows, balloon size represents support, and darkness represents lift.

III. TECHNICAL IMPLEMENTATION

A. Programming Language and Environment

This study was implemented in Python 3.12, a widely used programming language known for its extensive library ecosystem and active community support. All development and analysis were conducted in the Jupyter Notebook environment for its interactive coding, iterative analysis and documentation. Data was stored and accessed locally throughout the study.

Additional clustering analysis was conducted using JASP (jasp-stats.org) – an open-source statistical software by the University of Amsterdam.

B. Libraries

Data Manipulation: pandas, NumPy

Visualisation: matplotlib, seaborn

Clustering: scikit-learn - KMeans, MinMaxScaler, RobustScaler, silhouette_score, davies_bouldin_score

Association Rule Mining: mlxtend - apriori, fpgrowth, association_rules, TransactionEncoder

C. Methodology applied

Analytical Framework: CRISP-DM.

Models Implemented: K-Means Clustering (K-Means and K-Means++ initialization), Apriori Algorithm, FP-Growth Algorithm.

IV. METHODOLOGY

This study follows the **CRISP-DM** methodology – a widely adopted methodology for data mining and analytics projects. It is a six-phase iterative process, including business understanding, data understanding, data preparation, modelling phase, model evaluation, and deployment. This methodology is often adapted based on domain-specific requirements (Plotnikova *et al.*, 2022):

Figure 1 shows a diagram representing each step followed in this study. Phase 1 covers customer segmentation pipeline, from business understanding to cluster profiling, including RFM feature engineering, outlier treatment, feature scaling, and K-Means clustering with iterative preprocessing configuration evaluation. Phase 2 covers market basket analysis pipeline, starting with basket construction and transaction encoder per customer segment dataset, followed by Apriori and FP-Growth algorithms and the evaluation of association rules through support, confidence, and lift. The last step presents the findings after applying Market Basket Analysis to each customer segment.

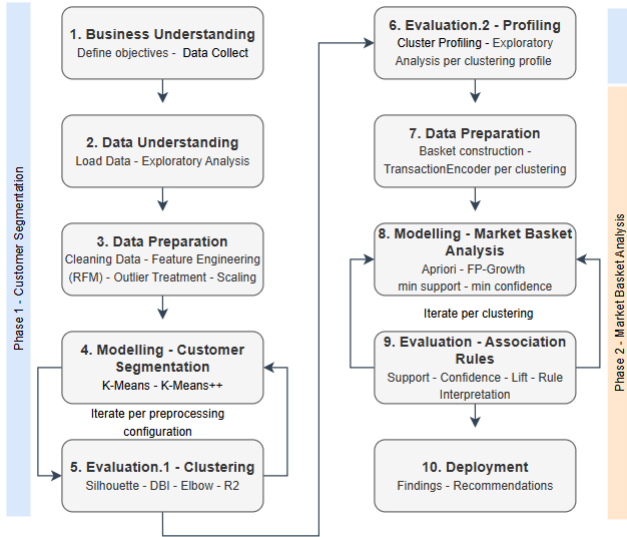


Figure 1- CRISP-DM pipeline diagram for customer segmentation and Market Basket Analysis.

A. Business Understanding

Traditional Market Basket Analysis assumes homogeneous customer behaviour, which may reduce the effectiveness of recommendations and cross-selling strategies. Applying customer segmentation to a transactional customer dataset prior to MBA enables more personalised product analysis and reveals segment-specific purchasing patterns aligned with the characteristics of each customer type.

B. Data Understanding

The Online Retail II (Chen, 2019), available in xlsx format, contains real transactional data from an UK-based online retail company operating between 2009 and 2011. The company sells unique all-occasion giftware, with a significant proportion of wholesale customers. The dataset was sourced from the UCI Machine Learning Repository, and it contains two sheets, one for 2009-2010, and the other for 2010-2011. The analysis was conducted on one year of data (2010-2011) which contains 541910 row records and 8 columns attributes.

The dataset contains five categorical features (*InvoiceNo*, *StockCode*, *Description*, *CustomerID*, *Country*), two numerical features (*Quantity* and *UnitPrice*) and datetime feature (*InvoiceDate*).

UnitPrice and *Quantity* were used to calculate *TotalPrice*, which together with *InvoiceNo*, *InvoiceDate* and *CustomerID* were used to create the *Recency*, *Frequency*, and *Monetary* features. *CustomerID* served as the unique identifier for

customer segmentation, while *InvoiceNo* was used as the transactional unit for Market Basket Analysis.

Table 3 - Dataset overview - column names and data types

```

RangeIndex: 541910 entries, 0 to 541909
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   Invoice          541910 non-null object  
1   StockCode       541910 non-null object  
2   Description     540456 non-null object  
3   Quantity        541910 non-null int64  
4   InvoiceDate     541910 non-null datetime64[us]
5   Price           541910 non-null float64  
6   Customer ID    406830 non-null float64  
7   Country         541910 non-null str

```

An initial exploratory examination of the dataset was conducted to assess data quality. Missing values, invalid entries, and outliers were identified after descriptive statistics and visual exploration. Table A.1 (Appendix A) shows missing values in the *CustomerID* column, negative values and outliers in the *Quantity* and *Price* columns. Tables A.2 and A.3 (Appendix A) reveal inconsistent names for *StockCode* and *InvoiceNo*. Table A.4 (Appendix A) indicates duplicates on *StockCode* and *Description* – ideally both should be unique per *InvoiceNo* and consistent in length. Data cleaning is an essential step for ensuring data quality; the following section details the steps taken to resolve these issues.

C. Data Preparation

1) Data Cleaning

Each step adopted in the cleaning and preparation phases is detailed in Table 4.

Table 4 - Cleaning and preprocessing steps.

STEP	ISSUES	ACTION TAKEN	RESULT
Missing Values	<i>Description</i> 0.0027% <i>CustomerID</i> 0.2493%	<i>CustomerID</i> missing values were removed	No more missing values 75% of the initial dataset
Invalid <i>InvoiceNo</i> and <i>Quantity</i> negative values	<i>InvoiceNo</i> prefixed with 'C' identified as cancellations and quantity negative values.	All transactions with <i>InvoiceNo</i> with 'C' were removed	Negative values solved 8900 values were dropped
Invalid <i>StockCode</i>	Values like 'POST', 'M', 'CZ', 'DOT', 'BANK CHARGES', 'PADS' were not following <i>StockCode</i> pattern	All patterns that are not five numbers or five numbers - letter were removed	Keeping just valid products 1555 values were dropped
Invalid <i>Description</i>	Not valid product name: 'Next Day Carriage'	All 'Next Day Carriage' were removed	Keeping just valid products 79 values were dropped
Invalid <i>Country</i> Value	Not a real country: 'Unspecified'	No action taken as does not affect the analysis	Not action taken
Invalid <i>UnitPrice</i>	<i>UnitPrice</i> less than 0.01	Values dropped	316 values were dropped
Extreme Outliers	Extreme outliers were displayed using a boxplot on <i>UnitPrice</i> and <i>Quantity</i>	Extreme values dropped for its small number of entries	4 values were dropped
Inconsistent <i>StockCode</i> and <i>Description</i>	<i>StockCode</i> and <i>Description</i> had inconsistent lengths - some codes contained multiple descriptions and vice versa	<i>StockCode</i> was dropped as <i>Description</i> was considered the more reliable product identifier	<i>StockCode</i> column was dropped
Duplicated <i>Description</i> per <i>InvoiceNo</i>	Duplicated <i>Description</i> entries within the same <i>InvoiceNo</i> were identified, caused by different <i>StockCode</i> , <i>Quantity</i> and <i>UnitPrice</i> for the same product	<i>Quantities</i> were summed and <i>UnitPrice</i> was averaged per <i>Description</i> and <i>InvoiceNo</i>	5002 duplicates were dropped 71.18% of the initial dataset
Feature Engineering	<i>Quantity</i> and <i>UnitPrice</i> merge	A new column was created multiplying unit price and quantity	<i>TotalPrice</i> column

Figures A.1 and A.2 in Appendix A illustrate a boxplot before and after the extreme outliers were dropped, a step noted in Table 4. Visualisation plots were created to better understand orders by product, customers, countries and sales throughout the year. Figures A.3 and A.4 illustrate total orders and revenue by month, and the top 10 products purchased by quantity and by revenue. Figures A.5 and A.6 presenting customer and country information can be seen in Appendix A.

Table 5 - Dataset overview - final configuration.

```

RangeIndex: 385734 entries, 0 to 385733
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   InvoiceNo         385734 non-null str
1   Description       385734 non-null str
2   InvoiceDate       385734 non-null datetime64[s]
3   CustomerID       385734 non-null int32
4   Country          385734 non-null str
5   Quantity         385734 non-null int64
6   UnitPrice        385734 non-null float64
7   TransactionID    385734 non-null str
8   TotalPrice       385734 non-null float64

```

Table 6 - Cleaned dataset sample.

InvoiceNo	Description	InvoiceDate	CustomerID	Country	Quantity	UnitPrice	TransactionID	TotalPrice
0 536365	CREAM CUPID HEARTS COAT HANGER	2010-12-01	17850	United Kingdom	8	2.75	536365_000001	22.00
1 536365	GLASS STAR FROSTED T-LIGHT HOLDER	2010-12-01	17850	United Kingdom	6	4.25	536365_000002	25.50
2 536365	KNITTED UNION FLAG HOT WATER BOTTLE	2010-12-01	17850	United Kingdom	6	3.39	536365_000003	20.34
3 536365	RED WOOLLY HOTTIE WHITE HEART.	2010-12-01	17850	United Kingdom	6	3.39	536365_000004	20.34
4 536365	SET 7 BABUSHKA NESTING BOXES	2010-12-01	17850	United Kingdom	2	7.65	536365_000005	15.30
...	...	...	...	...	...	...	...	...
385729 581587	CIRCUS PARADE LUNCH BOX	2011-12-09	12680	France	12	1.95	581587_385730	23.40
385730 581587	PACK OF 20 SPACEBOY NAPKINS	2011-12-09	12680	France	12	0.65	581587_385731	16.20
385731 581587	PLASTERS IN TIN CIRCUS PARADE	2011-12-09	12680	France	12	1.65	581587_385732	19.80
385732 581587	PLASTERS IN TIN STRONGMAN	2011-12-09	12680	France	12	1.65	581587_385733	19.80
385733 581587	SPACEBOY LUNCH BOX	2011-12-09	12680	France	12	1.95	581587_385734	23.40

Table 5 and 6 presents the final dataset configuration after cleaning. The next section presents a feature engineering process for creating a new dataset with Recency, Frequency, and Monetary values using the cleaned dataset.

2) Feature Engineering

a) Customer Segmentation

Following the methodology proposed by trentpark8800 (2024), the *Recency*, *Frequency*, and *Monetary* (RFM) features were created from the column *TotalPrice*, *InvoiceNo* and *InvoiceDate* aggregated by *CustomerID*. Monetary is the total amount spent (pounds) by a customer across all transactions, frequency is the number of total orders made by a customer, and recency is calculated as the time interval (days) between the customer's last purchase and the last day in the observation period. Table A.5 (Appendix A) shows the code applied to create the RFM features. Table 7 below shows the RFM dataset result.

Table 7 - RFM feature dataset sample.

	CustomerID	Monetary	Frequency	LastInvoice	Recency
0	12347	4310.00	7	2011-12-07	2
1	12348	1462.20	4	2011-09-25	75
2	12349	1457.55	1	2011-11-21	18
3	12350	294.40	1	2011-02-02	310
4	12352	1385.74	7	2011-11-03	36

The boxplot on figure 2 illustrates the distribution of customers by last purchase, the number of orders and total money spent. Outliers were detected in all three features, but only *Frequency* and *Monetary* distributions are affected by outliers. A histogram illustrating the distribution on RFM features is presented in Figure A.7 (Appendix A), providing a complementary visual representation on the spread of each variable.

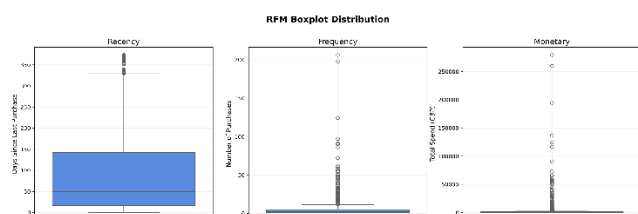


Figure 2 - RFM distribution before outlier treatment.

The methodology applied as a baseline for this study on K-Means application was proposed by Ling and Weiling (2025). The authors prepared the data by applying a robust technique based on IQR to eliminate outliers, and a min-max normalisation to standardize RFM features. IQR outlier removal was applied on Frequency and Monetary features, and a total of 469 customers were removed. Figure 3 shows how the boxplot distribution has improved.

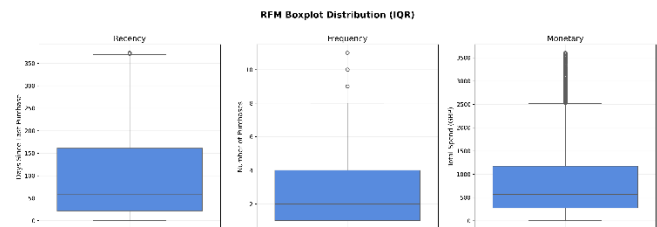


Figure 3 - RFM distribution after IQR outlier removal.

Transactional datasets are highly skewed, which makes outlier treatment and feature scaling necessary. The choices for IQR and Min-Max scaling are well established and used in research papers. Even though, K-Means is sensitive to outliers, an additional trimming technique was applied at the 99th percentile to preserve customer records while reducing influence of extreme outliers. Both methods were tested using K-Means application. Figure 4 shows the distribution after applying the trimming technique with a total of 63 customers removed.

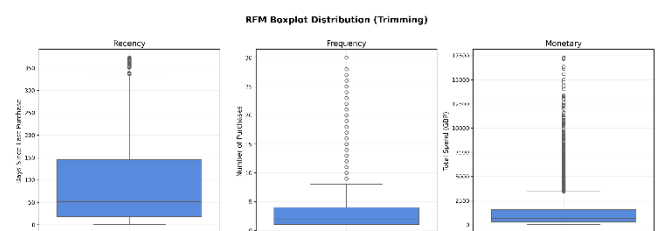


Figure 4 - RFM distribution after percentile trimming.

Robust scaling was examined as an alternative to Min-Max scaling for its reduced sensitivity to extreme values. Each scaling method was applied to three dataset configurations – the original RFM dataset, the IQR-treated dataset, and the trimmed dataset – to assess the influence of outlier treatment on feature distribution.

b) Market Basket Analysis

After clustering, clusters were assigned to the customer dataset (*Cluster_Label*). Following this, manual refinement adjusted the cluster configuration and each were profiled and named as a customer segment, then joined to the transactional dataset for the Market Basket Analysis.

Table 8 - Cluster labels assigned to the transactional dataset.

Cluster_Label	
Occasional	158531
Champions	112615
VIP	55218
Inactive	28817
New	16405
Lost	13841

A basket list was created aggregating each product purchased by order. *TransactionEncoder* was applied to transform the lists into a binary matrix, as Apriori expects

data in a one-hot encoded pandas DataFrame (Raschka). The binary matrix data is presented in Table 9.

Table 9 - Binary matrix sample after Transaction Encoder.

	10 COLOUR SPACEBOY PEN	12 COLOURED PARTY BALLOONS	12 DAISY PEGS WOOD BOX	12 EGG HOUSE PAINTED WOOD	12 HANGING ROSE EGGS HAND PAINTED	12 IVORY ROSE PEG PLACE SETTINGS	12 MESSAGE CARDS WITH ENVELOPES
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False

D. Modelling

1) Customer Segmentation

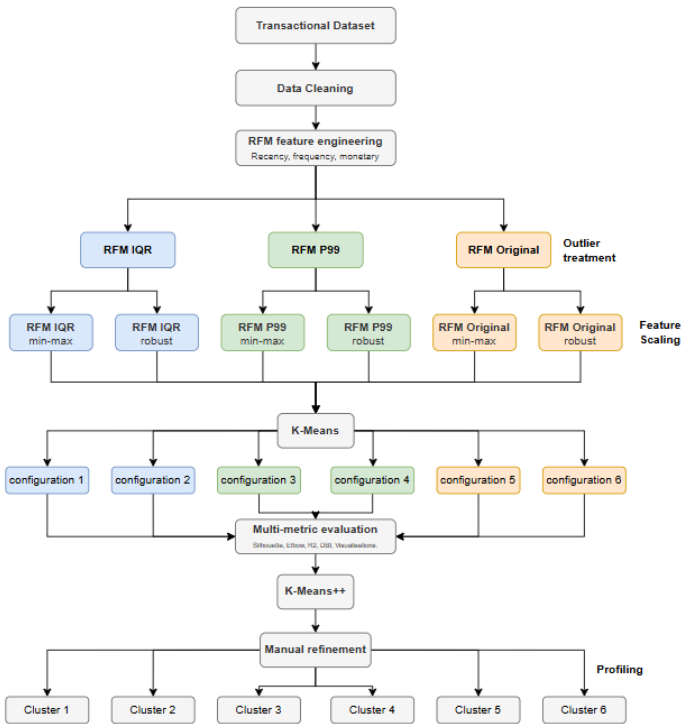
In the first phase of this study a clustering algorithm was applied to identify the optimal number of clusters and profile the resulting clusters based on customer RFM characteristics. An iterative experimentation approach was adopted across different dimensions: outlier treatment and scaling methods.

K-Means was applied using *KMeans* — *scikit-learn 1.8.0 documentation* across each preprocessed configuration, allowing a comparison of different outlier treatment and scaling methods. A K-Means++ initialisation was applied in the chosen model to evaluate whether any improvement in clustering results was produced. Table 10 presents all preprocessing and models conducted in this study.

Table 10 - Outlier, feature engineering and clustering methodology summary.

STEP	HOW IT WORKS	FORMULA	NOTES
OUTLIER TREATMENT			
Interquartile range (IQR)	Identifies outliers as values that fall below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$, where $Q1$ and $Q3$ represent the 25th and 75th percentiles respectively.	Lower = $Q1 - 1.5 \times IQR$ Upper = $Q3 + 1.5 \times IQR$ $IQR = Q3 - Q1$	It is a robust and well established outlier removal technique, but it can remove a significant amount of outliers that are common and important on a customer data.
Percentile capping (Trimming)	Removes extreme values that fall above the 99th percentile.	-	It still keeps data volume, but does not remove many outliers that can influence the model application.
FEATURE SCALING			
Min-Max scaling	Transform features by rescaling each feature to a [0,1] range.	$X_{scaled} = (X - X_{min}) / (X_{max} - X_{min})$	Maintain original distribution shape. Does not reduce the effect of outliers, but it is less sensitive than standard scaling.
Robust Scaling	It removes the median and scales using the quantile range (IQR).	$X' = (X - median) / IQR$	It reduces outlier sensitivity. It was tested, but the results were not better compared to Min-Max scaling.
CLUSTERING			
K-Means	The first iteration starts with random points as the initial centroids. Then, each data point is assigned to the closest centroid using the proximity measure chosen. Following the next iterations, the centroids change position until convergence criteria is met.	$J = \sum_{k=1}^K \sum_{x_i \in C_k} \ x_i - \mu_k\ ^2$	Scaling is required, it is sensitive to feature scaling. It is a well established method applied for clustering in many research papers.
K-Means++	Initialization method for K-means clustering it follows a weighted probabilistic score.	$P(x) = \frac{D(x)^2}{\sum_{x' \in X} D(x')^2}$	

Table 11 - Customer Segmentation pipeline



2) Market Basket Analysis

Phase II consisted of conducting the Market Basket Analysis using Apriori and FP-Growth algorithms. The transactional dataset was divided into smaller datasets using the customer segments identified in phase one. Each segment dataset was processed into basket format as described in section IV.C.2.b. Apriori and FP-Growth were conducted (*mlxtend*) with a minimum support threshold for each dataset, following the association rules with antecedents, consequents, confidence and lift evaluation metrics. These steps are illustrated in Figures 5 and 6.

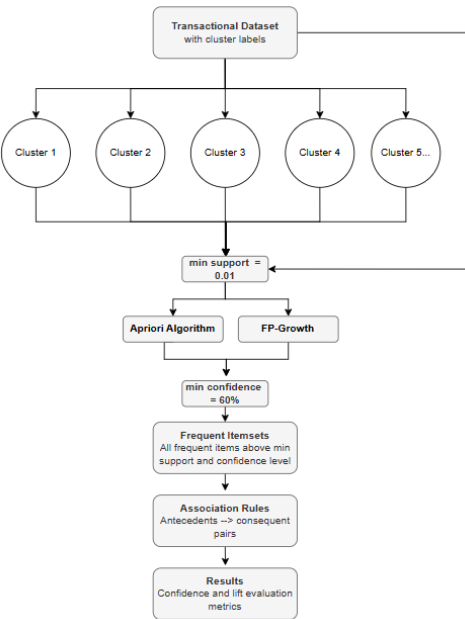


Figure 5 - Market Basket Analysis pipeline.

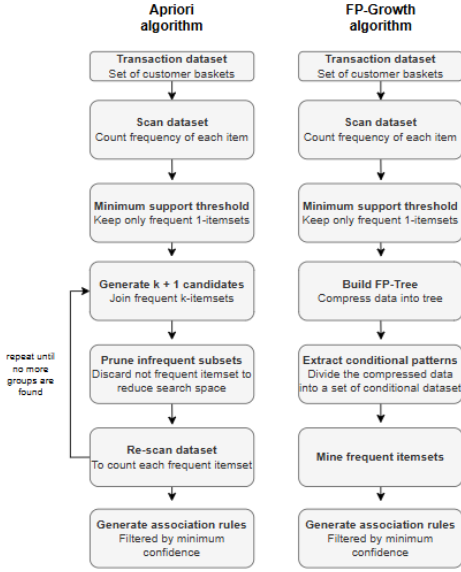


Figure 6 - Apriori and FP-Growth algorithm flowchart.

E. Evaluation

1) Clustering

Given the absence of ground truth in unsupervised learning, a multi-factor evaluation framework was adopted, consistent with the recommendation of Wasilewski (2024) who demonstrated that single-metric evaluation produces unreliable results. Table 12 summarises the seven evaluation criteria applied in this study, covering internal validity metrics, k selection methods, variance explanation, and qualitative cluster interpretation.

Table 12 - Clustering evaluation framework.

STEP	HOW IT WORKS	NOTES
Silhouette Score	Validates the clustering performance based on how similar its compared to the nearest neighbouring cluster (Rousseeuw, 1987).	Close to 1 is well separated. Close to 0 is overlapping. Helps to predict optimal k where the curve achieve higher values.
Elbow Method (WSS)	Plots the within-cluster sum of squared (WSS) against increasing values of k.	The optimal k is where the curve stops to meaningfully reducing within-cluster variance.
Davies-Bouldin Index	Evaluates clustering quality measuring how high each cluster is compared to how far the clusters are from each other (Davies and Bouldin, 1979).	Lower index values indicate better clustering, which means clusters are most distinct to each other.
R ² - Variance explained	Measure proportion of total variance in the dataset that is explained by the cluster structure.	From 0 to 1 / High is better. R ² increases naturally as the number of clusters grow, it does not mean its measnifnful.
Cluster Profile Analysis	Recency, Frequency, and Monetary values will be compared across the clusters to check if they have distinct characteristics.	Mean, Median evaluation
Cluster Size Balance	Examines the number of customers to each cluster to verify their overall balance.	Very low values, ~5% can have no meaninfull value for a customer segment analysis or even highly unequal distributions 90/10/5
3D Visual Inspection	A three-dimensional scatter plot on RFM features provides a visual validation to each cluster distribution.	A balanced, well distributed with not many overlapping data points are healthy results.

2) Association Rules

As noted in section II.2, support and confidence are the primary measures used to evaluate association rules, yet relying only on this framework can produce misleading

results. A rule may satisfy both thresholds whilst reflecting no genuine relationship between itemsets. To address this limitation, lift was chosen as an additional correlation measure for a more robust evaluation of the strength of association between itemsets. It is important to note that association rule evaluation can be approached both subjectively, where domain expertise is required to assess context relevance, and objectively, where statistical measures are used to assess rule strength, as in this study (Han, Kamber and Pei, 2012). Table 13 presents the three evaluation measures applied with their definitions, formulas, and the thresholds chosen for this study. Additionally, as noted in Section II.2, bar plot, scatterplot and heatmap were applied to complement statistical evaluation of association rules, enabling a visual exploration of rule distribution, metric relationships and product patterns across customer segments.

Table 13 - Association rule evaluation measures.

EVALUATION METHODS	DEFINITION	NOTES	FORMULA
Support	Measures how frequent an itemset appears across all transactions, expressed as a proportion of the total number of transactions (N)	A min_support of 0.01 was applied. An itemset that does not appear in at least 1% of the total transactions will be pruned out.	$support(A, B) = \frac{P(A \cup B)}{N}$
Confidence	The probability that B occurs given that A has occurred.	A min_confidence threshold of 60% was applied. An itemset with a confidence level lower than 60% will be pruned out.	$confidence(A, B) = \frac{P(A \cup B)}{P(A)}$
Lift	Measures the strenght of the association between itemsets. Indicating how many times more likely itemset B occurs when itemset A has occurred.	Lift = 1 (A and B are independent and not correlated). Lift > 1 (A and B are positively correlated. Customers who buy A are more likely to buy B). Lift < 1 (A and B are negatively correlated. Customers who buy A are less likely to buy B). Evaluation criterion: > 1 indicates a positive association.	$lift(A, B) = \frac{P(A \cup B)}{P(A) P(B)}$

F. Deployment

Following the evaluation of both Customer Segmentation and Market Basket Analysis, the findings were interpreted based on their analytical value. This study evaluated whether the combined pipeline of clustering followed by MBA produces more segment-specific and interpretable association rules than MBA applied to the full customer dataset.

The phase one consisted of applying the final K-Means cluster assignments to the transactional dataset. Each segment was processed through the Apriori algorithm to generate association rules per segment. The resulting rules were evaluated using support, confidence and lift, and compared against the full dataset baseline to assess whether the segmented pipeline reveals additional and more targeted purchasing patterns. The findings are presented in section V.

V. RESULTS AND EVALUATION

This section presents the results of the customer segmentation analysis and market basket analysis. Customer segmentation is organized across five stages: cluster configuration evaluation, performance selection, manual refinement, cluster profiling and business metrics by customer segment. Market basket analysis is organized across four stages: MBA on the full dataset, Frequent itemset and rule generation by segment, association rule evaluation by segment, segment vs full dataset comparison.

A. Customer Segmentation

1) Cluster Configuration Evaluation

K-Means was applied to different preprocessing configurations on RFM features as described in section IV.D – Table 10. Table 14 presents the clustering performance metrics obtained across all configurations.

Table 14 - Clustering configuration evaluation.

Configuration	Scaler	Outlier Method	k	Silhouette	DBI	R2	Count	Distribution (%)
C1 (Baseline)	MinMax	IQR	3	0.48	0.753	0.705	3864	53/25/22
C2	Robust	IQR	3	0.45	0.768	0.687	3864	53/26/21
C3	MinMax	None	3	0.65	0.501	0.875	4333	68/18/14
C4	Robust	None	4	0.75	0.556	0.848	4333	96/3.5/0.4/0.1
C5.1 ✓	MinMax	q99	4	0.51	0.417	0.743	4270	65/13/12/10
C5.2	MinMax	q99	3	0.56	0.691	0.815	4270	65/25/10
C6	Robust	q99	4	0.45	0.768	0.756	4270	55/24/17/4

As Wasilewski (2024) noted, cluster selection based on a single metric may lead to inconsistent outcomes — a pattern confirmed here, where C4 and C5.1 each rank highest across different metrics.

As shown in Table 14, outlier treatment produced an improvement in distribution across clusters, but it reduces the amount of data for the analysis. Robust scaling did not offer any meaningful results; it performed slightly worse than min-max scaling. No single internal validity metric consistently showed an optimal number of clusters across all tested configurations. Therefore, cluster distribution balance, visual inspection of the RFM features, and the context of this study were required to determine the optimal k value.

Overall, C1 (baseline) and C2 demonstrated the most balanced cluster distribution, but their metric scores were the lowest across configurations. C3 and C4 achieved the highest scores; however, C4 has a highly unbalanced cluster size distribution. C6 is outperformed by C5.1 and C5.2, both of which emerged as the most promising configurations with balanced metric scores and cluster distribution values. The selection of the final configuration is discussed in the following section.

2) Performance Selection

The selection was based on a combination of criteria, not one single metric. Visualisation plots and project context were important to compare results across configurations. Table 14 reveals that min-max scaling outperformed robust scaling across metrics. Consequently, C1, C3, C5.1 and C5.2 were retained for further evaluation.

To support the configuration selection, RFM mean profiles were observed visually for each configuration (Figure 7; Figures B.1, B.2 (Appendix B)). Figure 7 illustrates the RFM mean profiles for C3, identifying that Frequency and Monetary means across all clusters are almost zero, this demonstrates how extreme outliers can impact on K-Means clustering. C3 configuration is mostly uninterpretable. On other hand, Figures B.1 and B.2 (Appendix B) illustrate that each cluster has different patterns, which is expected to a well-defined segmentation.

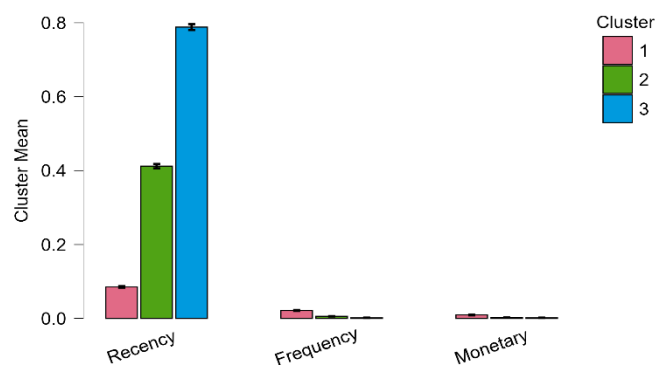


Figure 7 - RFM mean distribution for C3 configuration.

Comparing Figures B.3, B.4 and B.5 (Appendix B), C1 (baseline) demonstrates a more balanced and defined cluster distribution across RFM features, different from the C5.1 and C5.2 plot. This illustrates how the presence of outliers influences the distribution of data across RFM features. Even though C1 is more balanced, C5.1 and C5.2 retain more customer information.

The configuration C5 has two variants with k=3 and k=4, since both have overall good results and could be selected as the main configuration. As seen in Figure 8 and 9, Elbow and Silhouette results were acceptable for both cluster solutions. C5.2 has a higher silhouette, C5.1 a better DBI, and R2 is good for both. The transition from k=3 to k=4 maintained the overall cluster distribution, with one cluster subdivided into two smaller groups (Figures B.4 and B.5 (Appendix B)).

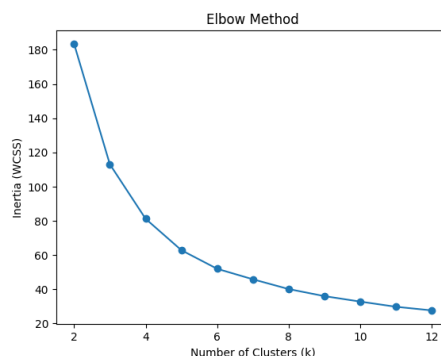


Figure 8 - Elbow method - within-cluster sum of squares for C5.1 configuration.

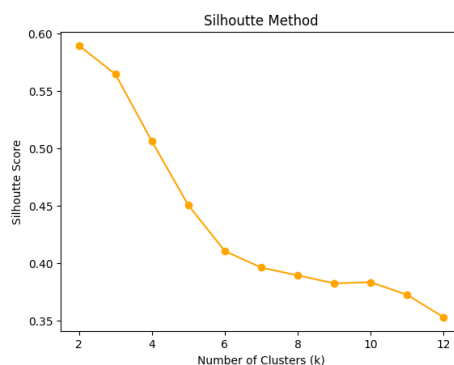


Figure 9 - Silhouette score across k values for C5.1 configuration.

A three-cluster solution provides a good segmentation structure, but it can be insufficient to capture the behavioural diversity presented in a customer database. Consequently, C5.1 was selected as the final configuration, as it demonstrated good metric scores, an acceptable cluster size distribution, clear spatial separation, and distinct mean RFM profiles across all clusters. The final customer segmentation contains the following configuration:

- Cluster 0 = 2490 customers,
- Cluster 1 = 770 customers,
- Cluster 2 = 401 customers,
- Cluster 3 = 609 customers.

The means, median and customer distribution can be accessed in Table B.1 (Appendix B).

3) Manual Refinement

Prior to cluster profiling, a visual and analytical inspection presented an opportunity for refinement. As customer behaviour is complex and segmentation objectives vary across use cases, contextual knowledge of the data can be valuable for interpreting and improving cluster solutions. Two refinements were identified: First, the extreme outliers removed during preprocessing exhibited a distinct purchasing pattern and they were reintegrated as a new cluster segment (Figure B.6 in Appendix B). Second, within Cluster 0, a subgroup of customers with low Recency but a Frequency of exactly one was identified. These customers had made only one purchase and made it recently – a behavioural profile distinct from the rest of Cluster 0, whose customers had already made repeat purchase. This subgroup was split out as its own segment representing new customers rather than an established purchasing pattern.

Following both refinements, Cluster 0 was split into two: Cluster 0 with 1831 customers, and Cluster 5 with 659 customers. The extreme outliers have become Cluster 4 with 61 customers. The final configuration before cluster profiling was:

Cluster 0 = 1831 customers,
Cluster 1 = 770 customers,
Cluster 2 = 401 customers,
Cluster 3 = 609 customers,
Cluster 4 = 61 customers,
Cluster 5 = 659 customers.

Means, median and customer distribution can be accessed in Table B.2 (Appendix B).

4) Cluster Profiling

In this section the RFM profiles of each cluster were examined to determine meaningful customer characteristics and assign appropriate segment labels. As shown in Figures 10 and 11, considerable patterns were observed across cluster segments in all three RFM dimensions, suggesting a successful clustering algorithm. Figure 10 illustrates the average RFM features by cluster, and Figure 11 shows the distribution of each cluster from the boxplot. Cluster 4 (extreme values) was removed from boxplot since it would influence the visualisation of the other clusters.

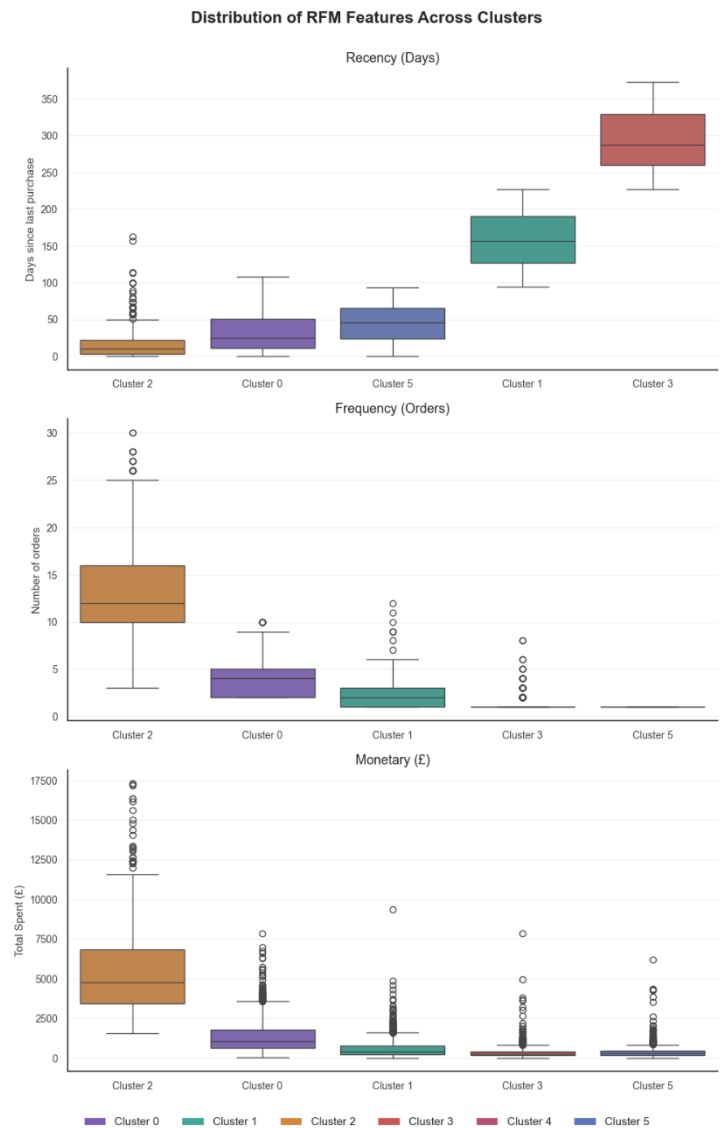


Figure 10 - RFM distribution across clusters.

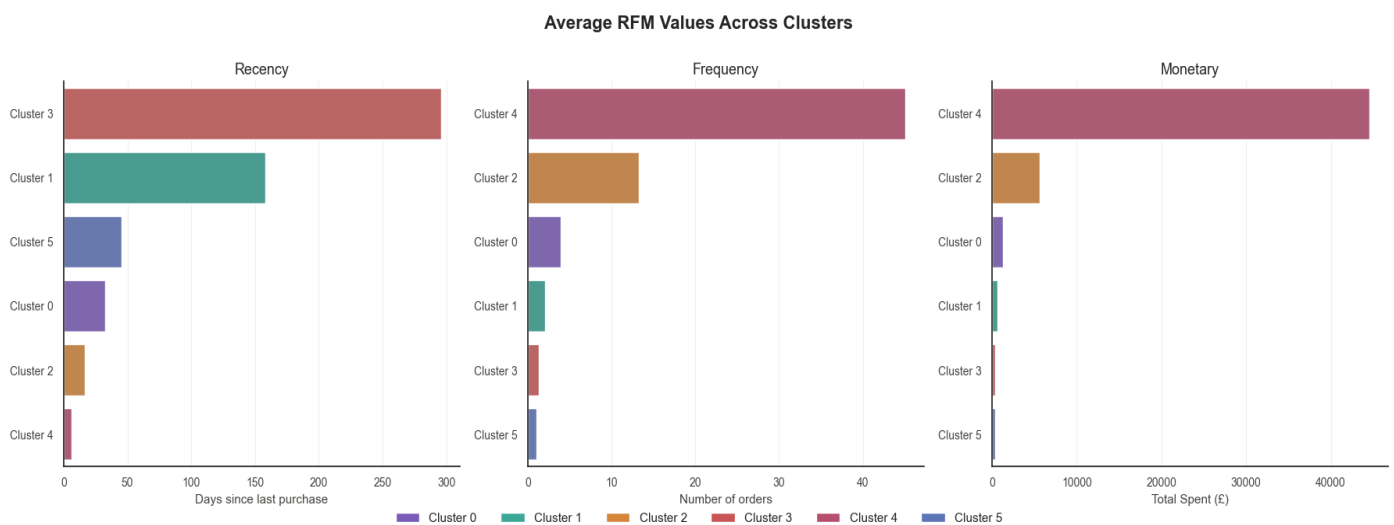


Figure 11 - Average RFM values by segment after manual refinement.

Cluster 0 (Occasional): It shows low recency average and the third highest frequency and monetary, indicating these customers purchased within the last month on average, with approximately 4 orders per year and a monetary average of £1,300. This suggests an occasional buyer, who remain active, but has mostly average values compared to other clusters.

Cluster 1 (Inactive): It presents the second highest recency, low purchase frequency and a low monetary value, indicating that these customers have spent, on average, half of Cluster 0, having made an average of two orders, and an average last purchase of approximately five months ago. This suggests that despite having been inactive for some time, still has more frequency and monetary value than Cluster 3.

Cluster 2 (Champions): Excluding Cluster 4, Cluster 2 contains the highest monetary and frequency values, while the lowest recency with an average of two weeks for the last order, but with some outliers. As one of the most active and high-value segments, Champions represent a critical customer group.

Cluster 3 (Lost): This segment presents similar frequency and monetary value with Cluster 5, but with a contrast in recency. Overall, their frequency and monetary values are one of the lowest, having the highest recency with an average of approximately one purchase, this indicates a predominantly one-time buyer. As their last purchase was almost 9 months ago, on a one-year dataset, this segment is considered Lost.

3D RFM distribution by cluster (VIP not included)

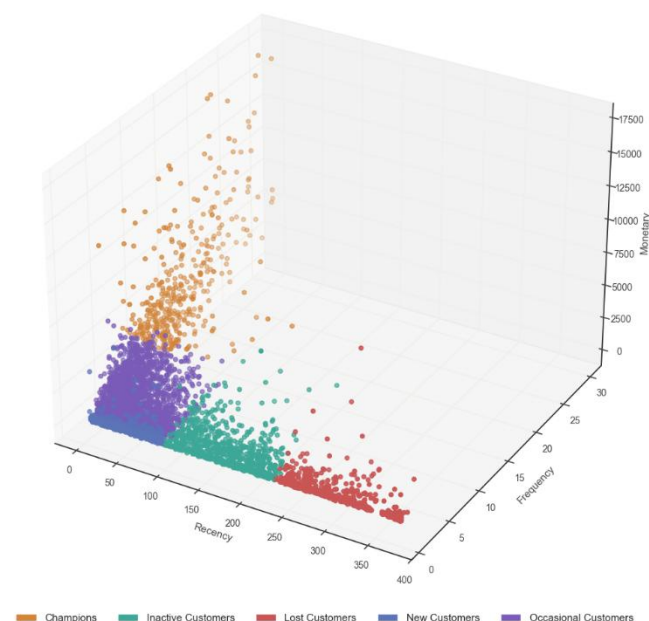


Figure 12 - Three-dimensional RFM by cluster - VIP excluded.

Cluster 4 (VIP): Despite representing a small number of customers, this segment the lowest average of recency and with monetary and frequency averages much higher than Cluster 2. VIP segment contributes disproportionately to overall revenue and order frequency, despite representing a small proportion of the customer base.

Cluster 5 (New): As previously noted, this segment contains similar monetary and frequency values compared to

Cluster 3. The key distinction is recency, with a frequency value of 1 and recency average of 45 days, this indicates that these customers have made their only and last purchase recently, consistent with a new customer profile.

In summary, the clustering procedure indicated six distinct customer segments, each exhibiting a clearly differentiated RFM profile. Table B.3 (Appendix B) presents more descriptive statistics on RFM values for each segment, while Figures 12 and 13 illustrate the distribution of RFM features across clusters in a three-dimensional feature space.

3D RFM distribution by cluster

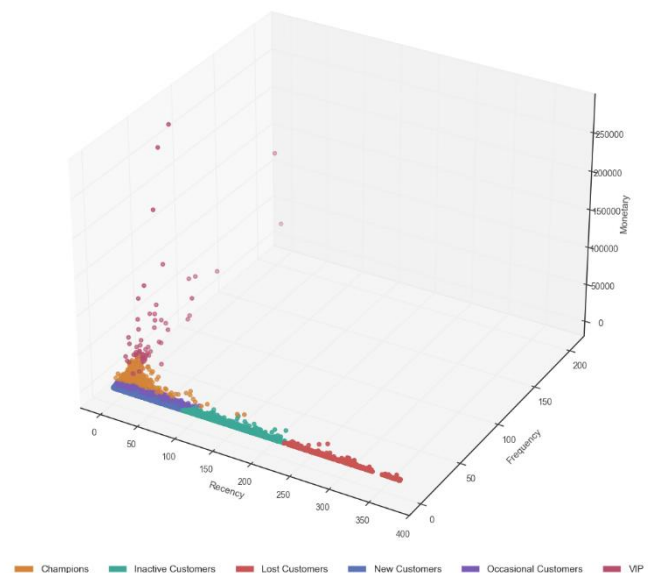


Figure 13 - Three-dimensional RFM by cluster - VIP included.

These findings address RQ1, confirming that RFM-based K-Means clustering can be effectively applied to identify distinct customer segments, each defined by contrasting behavioural profiles across recency, frequency and monetary dimensions.

5) Business Metrics by Customer Segment

Ultimately, cluster profiles were assigned to the transactional dataset, enabling an exploratory analysis of key business metrics – including order volume, revenue, line item – across the identified segments.

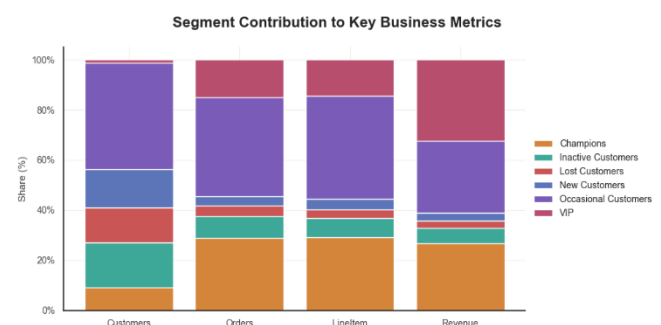


Figure 14 - Segment contribution to key business metrics - proportional stacked bar chart.

Figure 14 presents the proportional contribution of each segment across four key business metrics: customer count, order volume, total line items, and revenue. Occasional segment represents one of the largest groups for each key

metric. New and Lost segments recorded the lowest values across three of four metrics. VIP and Champions have the smallest customer base, but with a high number of orders, line items, and the largest share of total revenue. Inactive segment counts are the second largest, but with low orders, line items, and revenue.

It is worth noting that segments vary in size, ranging from more than 5000 orders (Occasional and Champions segments) to fewer than 1000 orders (Lost and New segments). Segments with smaller datasets may produce limited results for association rule interpretation.

These findings demonstrate that RFM-based features and K-means clustering provide an effective framework for translating raw transactional data into customer profiles. The following section presents the results of the Market Basket Analysis, applying association rules to each of the six segments and examining whether this approach can identify product patterns in relation to behavioural characteristics of each group.

B. Market Basket Analysis

1) MBA on the full dataset

Apriori and FP-Growth algorithms were first applied to the full dataset using the thresholds defined in Section IV.E. FP-Growth was faster (1.9550s) than Apriori (5.3417s) in execution time. Apriori and FP-Growth produced identical frequent itemsets, while FP-Growth achieved a shorter execution time in the benchmark. Apriori was retained for the subsequent segmented analysis for methodological consistency.

Table 15 - MBA summary table: baseline /full dataset.

Segment	Total Transactions	Total Products	Frequent Itemsets	Rules	Avg Support	Avg Confidence	Avg Lift	Itemsets
Full Dataset	18358	3845	997	162	0.0139	0.699	22.8231	1 = 623 items 2 = 309 items 3 = 62 items 4 = 3 items

The full dataset contains 18358 unique orders and 3845 unique products. Using a minimum support threshold of 0.01, the Apriori algorithm identified 997 frequent itemsets, indicating that these itemsets appear in at least 1% of all transactions. Table C.1 (Appendix C) shows the 10 most frequent itemsets. While single itemsets do not generate association rules directly, they serve as building blocks for candidate generation and provide insight into the most frequently purchased individual items.

Association rules were generated using a minimum confidence threshold of 0.60, resulting in 162 rules that satisfied this criterion. The average support of these rules is 0.0139, suggesting that most rules are close to the threshold and appear in approximately 1% of all transactions. The average confidence of 0.699 indicates that, when the antecedent itemset occurs, the consequent itemset follows with a probability of 69%. The average lift of 22.8231 is notably high, indicating that when an antecedent itemset occurs, the consequent itemset is ~22 times more likely to occur than would be expected by chance, representing a strong association between itemsets.

2) Frequent Itemset and Rule Generation by Segment

Table 16 presents the frequent itemsets and association rules generated by the Apriori algorithm across all six customer segments, using the same threshold adopted to the full dataset.

Table 16 - MBA summary table: all segments.

Segment	Total Transactions	Total Products	Frequent Itemsets	Rules	Avg Support	Avg Confidence	Avg Lift	Itemsets
Occasional	7259	3540	1004	146	0.0136	0.7098	26.3826	1 = 679 items 2 = 281 items 3 = 43 items 4 = 1 items
Champions	5333	3358	1789	994	0.0131	0.7141	13.6984	1 = 612 items 2 = 684 items 3 = 383 items 4 = 100 items 5 = 10 items
VIP	2749	3201	1137	308	0.0142	0.779	26.1158	1 = 582 items 2 = 244 items 3 = 38 items 4 = 3 items
Inactive	1570	2852	867	240	0.0128	0.7815	43.4474	1 = 596 items 2 = 459 items 3 = 94 items 4 = 8 items
New	659	2419	1774	573	0.0138	0.778	25.6235	1 = 823 items 2 = 761 items 3 = 142 items 4 = 19 items
Lost	788	2273	963	276	0.0134	0.8038	43.7957	1 = 595 items 2 = 325 items 3 = 40 items 4 = 3 items

In summary, it is possible to identify meaningful and strong patterns across each segment, ranging from 867 itemsets in the Inactive segment to 1789 in Champions, while the number of generated rules varies from 146 in Occasional to 994 in Champions. Although some segments contain more transactions, all segments produced more rules than Occasional segment.

Champions is the only segment containing itemsets of size five, while other segments are limited to itemsets of up to size four. As expected, Single itemsets do not contribute to rule generation but highlight frequently purchased individual items per segment.

An interesting observation is that New customers, despite having the lowest number of transactions, generated 1774 frequent itemsets and 573 association rules. This is one of the highest rule volumes among the segments and warrants further consideration when interpreting the segmented results. The evaluation of strength of association rules through support, confidence, and lift across each segment is discussed in the following section.

3) Association Rule Evaluation by Segment

Overall, the association rules generated across all six segments indicate meaningful purchasing patterns when evaluated through support, confidence, and lift. Average support values are close to the minimum threshold of 0.01 across all segments. Additionally, Association rules were generated using a minimum confidence threshold of 0.60, while lift was subsequently used to evaluate the strength of the resulting associations. The average confidence values across all segments exceed 0.70, surpassing the minimum confidence threshold of 0.60, suggesting that, on average, when an antecedent itemset is purchased, the consequent itemset follows in at least 70% of cases. the average lift values across segments range from 13.70 in the Champions segment to 43.80 in Lost, exceeding the threshold. Even the lowest average lift indicates that co-purchasing occurs approximately 13 times what is expected by chance, confirming genuine patterns and non-random associations.

Figure 15 illustrates the lift distribution across all segments, alongside the full dataset baseline. The Occasional, VIP, and New segments display distributions similar to the full dataset. Champions segment presents the lowest lift range, with high concentration of outliers. Inactive and Lost, despite being the smallest transaction volume, contain the

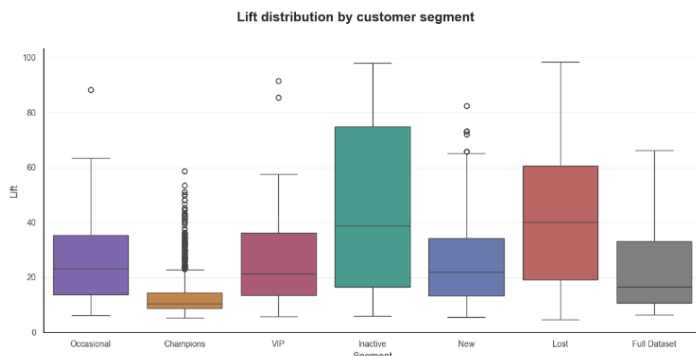


Figure 15 - Lift boxplot by segment.

association rules, with the Inactive and Lost segments reaching values close to 100.

Tables C.2 to C.7 in Appendix C present the top 10 association rules by lift for each segment, including antecedent and consequent itemsets alongside support, confidence, and lift values.

- Herb marker family is in most segments, excepting Champions and New customers. Inactive customers exclusively contain herb marker across all rules.
- Regency tableware is in most segments, excepting Lost and Inactive customers.
- Jam jar, party candles, and Poppy's playhouse

Support-Lift-Confidence Scatterplot by Segment

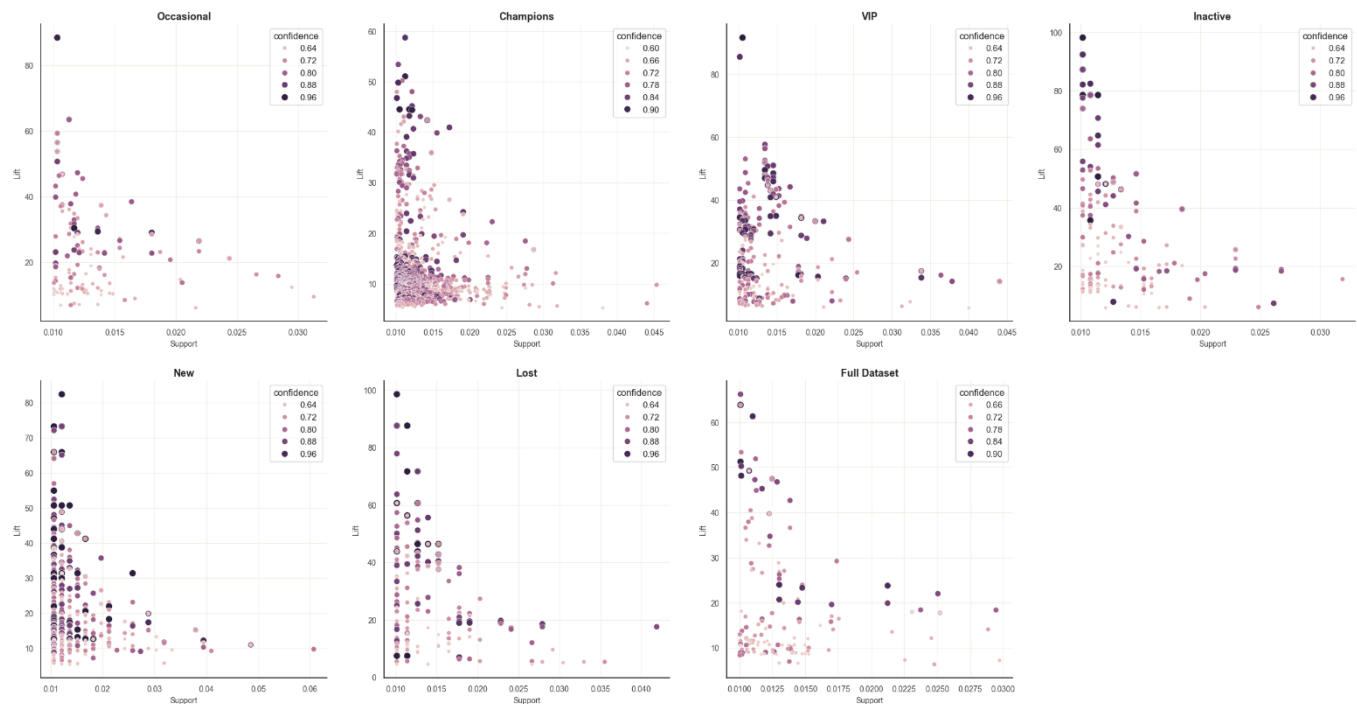


Figure 16 - Support, Lift and Confidence scatterplot.

highest mean and maximum lift values. Overall, all segments remain well above the threshold of 1, reinforcing the strength of the identified association rules and suggesting purchasing behaviour patterns rather than coincidence.

Figure 16 presents the distribution of support, lift, and confidence values across segments, providing a comprehensive view of the three-evaluation metrics. The Champions and New segments present the highest density of rules, followed by VIP, Lost, Inactive and Occasional. Despite having the largest number of transactions, the Occasional segment and the full dataset generated the lowest number of rules, suggesting that higher transaction volume does not necessarily produce stronger or more association rules.

Across all segments, support values are largely concentrated between 0.01 and 0.02, indicating that most co-purchased patterns occur in 1-2% of transactions. Confidence values are broadly distributed across segments, with average values exceeding 0.70, as reported in Table 16. Lift values for most rules fall between 10 to 40, confirming strong

variants are exclusive to Champions customers.

- Wooden decoration sets are exclusive to VIP customers.
- Coffee mugs, paper cups, and paper plates are exclusively in Occasional customers.
- Mug in Box is exclusive to New customers.
- Magic garden decoration pairs are exclusive to Lost customers.

These findings display the 10 strongest association rules (by lift) differ across segments. Even though some products like herb marker family and regency tableware appear across multiple segments, many itemsets are exclusive to a single customer segment. This highlights distinct and meaningful purchasing behaviour within each group and provides evidence that a segmented MBA approach reveals additional association rules.

These findings address RQ2, demonstrating that meaningful association rules with strong lift, confidence and support values can be discovered within each customer segment, confirming genuine co-purchase patterns across all six customer types. The following section compares the

outcomes of applying MBA at the customer-segment level versus the full dataset.

4) Segmented vs Full Dataset Comparison

This section addresses the third research question by evaluating whether applying MBA at the customer-segment level produces stronger and more distinct rules compared to the full dataset baseline.

Figure 17 illustrates the total number of rules generated for each segment alongside the full dataset. The graphic indicates that there is a substantial difference between rule volume produced across all segments compared to only 162 rules in the full dataset.

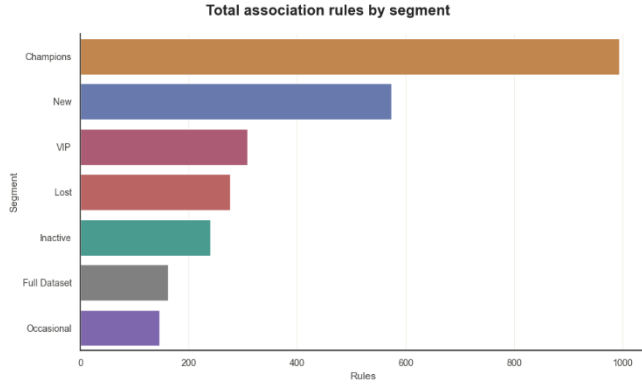


Figure 17 - Total association rules by segment.

As reported in Table 15 (Section V.B.1), the full dataset presents comparable average confidence, support and lift values to those observed in the segmented rules, suggesting that rule quality at the aggregate level is not necessarily weaker than at the segment level, as further illustrated in Figures 15 and 16. Additionally, Table C.1 (Appendix C) displays that items such as regency tableware, and poppy's playhouse sets dominate the top association rules in the full dataset, indicating that the full dataset demonstrates different association rules from those identified across each segment-level result.

Figure 18 illustrates a heatmap of the 10 most frequently purchased individual products by segment. Even though single itemsets do not generate association rules, they provide valuable insights into the most frequently purchased individual products. As presented in the heatmap, diverse purchasing patterns are identifiable across segments, including to the full dataset, indicating that it is possible to find different frequent items and supporting the presence of distinct product preferences by customer segment and full dataset. While some products appear consistently across multiple segments, others are exclusive to specific segments.

It is important to acknowledge that there is a limitation on segmented MBA. Segment-level analyses are based on smaller transaction volumes, ranging from 659 to 7259 transactions, compared to 18358 in the full dataset. As a result, the minimum support threshold of 0.01, while appropriate for large dataset, may be relatively low for smaller segments (particularly those with fewer than 1000 transactions). This can lead to an inflation in the volume of generated rules and metric values. Consequently, elevated rule volumes and metric values in smaller datasets do not

necessarily indicate better purchasing pattern strength compared to larger segments.

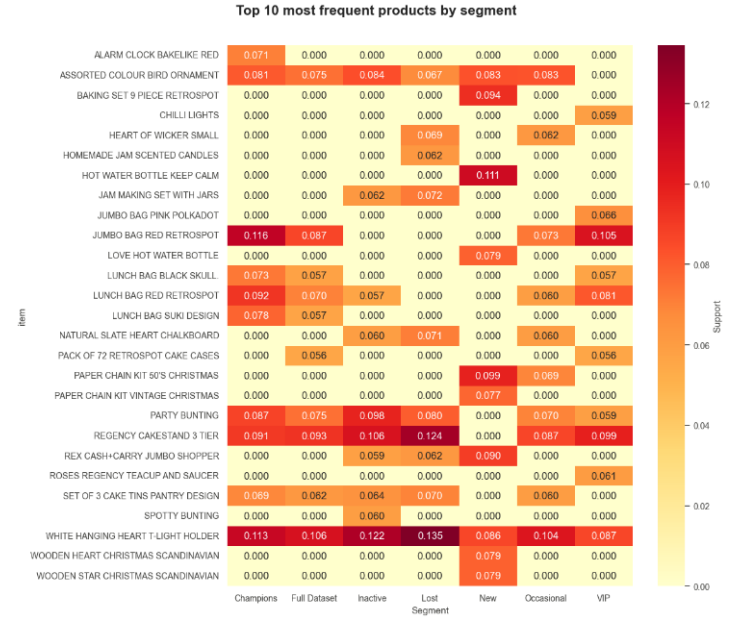


Figure 18 - Top 10 most frequent items by segment.

Overall, the comparison between segmented and full dataset MBA demonstrates that applying MBA at the segment level reveals a greater volume and diversity of association rules than aggregate analysis alone. Even though the segmented approach does not necessarily produce stronger quality rules, it still uncovers segment-specific purchasing patterns that are undetected in the full dataset, providing a more granular analysis of purchasing behaviours across customer segments. These findings support the proposed pipeline of K-Means customer segmentation followed by an MBA per segment using Apriori algorithm.

VI. CONCLUSION

A. Limitations

This study applied the proposed pipeline to a single retail transactional dataset; the findings may not generalise to other contexts. Although the dataset contains enough transactions, it includes only one year of activity, which may restrict the identification of long-term purchasing patterns and seasonal trends. Additionally, the customer segmentation was conducted exclusively using RFM features, excluding broader behavioural dimensions such as browsing activity, demographic characteristics and product engagement, which would enable a more refined customer profiling and improve segmentation quality.

K-Means is sensitive to outliers, which may still influence cluster assignments in transactional dataset where high-value customers influence the data distribution. Alternative clustering algorithms or other machine learning algorithms could be explored to assess whether alternative approaches provide more refined segmentation outcomes.

Many limitations are acknowledged in relation to MBA. A fixed minimum support threshold of 0.01 was applied across all segments to ensure comparability; however, for smaller datasets such as New, Inactive, and Lost customers,

this threshold produces lower absolute transaction count, which may reduce the statistical reliability of the rules generated. In a dataset with 800 transactions, 0.01 of support requires a product to appear in only 8 transactions to be considered frequent, which may be insufficient to confirm a real pattern. An adaptive threshold to segment size could address this limitation.

The threshold selection for minimum support, lift, and confidence was based on literature conventions rather than domain expert validation, and the same applied to association rules evaluation, clustering validation and profiling. Domain expertise would improve the contextual validity of the findings for both MBA and customer segmentation, since specific retail context was not available in this study.

B. Future Work

Considering the limitations, there are several directions for future research. First, applying the proposed pipeline to multiple retail datasets across different industries would strengthen the generalisability of the findings. Datasets containing longer periods of time and capturing different features beyond RFM could further enable a more comprehensive customer profiling and improve segmentation quality.

Future work could explore alternative clustering algorithms or deep learning representations to assess whether alternative approaches result in more refined and meaningful customer segments. Regarding the MBA phase, a more rigorous threshold selection process, applying an adaptive support threshold by segment size to assess whether there is any improvement in interpretability of the rules. Finally, applying the full pipeline in collaboration with a domain expert for cluster validation, segment profiling and association rule evaluation would improve the contextual validity of the findings and provide a more robust evaluation of the pipeline's practical applicability.

C. Conclusions

This study applied a two-phase data mining pipeline to a retail transactional dataset, combining RFM-based K-Means customer segmentation with Market Basket Analysis per segment using the Apriori algorithm, following the CRISP-DM methodology. The pipeline was proposed to evaluate whether applying MBA at the customer segment level produces stronger and more distinct association rules than aggregate analysis alone.

K-Means clustering applied to RFM features identified four distinct customer segments each containing different behavioural profiles in terms of recency, frequency, and monetary value. The final configuration of clustering algorithm was refined through manual informed adjustment, validated through a multi-metric evaluation framework producing the six final segments.

The Apriori algorithm generated meaningful association rules across all six customer segments, with lift values consistently above 1, and confidence average exceeding 0.70 across all segments. Analysis of the association rules revealed that even though same products appear across multiple segments, several products are exclusive to a single customer segment. This confirms that each segment shows distinct purchase patterns compared to each other.

The comparison between segmented and full dataset MBA demonstrated that, at customer level, MBA reveals a

greater volume and diversity of association rules than aggregated level alone. Although the segmented MBA did not demonstrate uniformly improved lift, confidence or support values relative to the full dataset, the segmented approach uncovered itemset rules that were undetected in the full dataset, confirming the analytical value of this pipeline.

Overall, this study provides evidence that integrating Customer Segmentation with Market Basket Analysis provides a more granular and informative analytical approach than an MBA applied to the full dataset, offering a methodological contribution to how e-commerce retailers could use this pipeline to better understand customer purchasing behaviours.

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A. Appendix A – Data Preprocessing

	Quantity	InvoiceDate	Price	Customer ID
count	541910.00	541910	541910.00	406830.00
mean	9.55	2011-07-04 13:35:22.342307	4.61	15287.68
min	-80995.00	2010-12-01 08:26:00	-11062.06	12346.00
25%	1.00	2011-03-28 11:34:00	1.25	13953.00
50%	3.00	2011-07-19 17:17:00	2.08	15152.00
75%	10.00	2011-10-19 11:27:00	4.13	16791.00
max	80995.00	2011-12-09 12:50:00	38970.00	18287.00
std	218.08	NaN	96.76	1713.60

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
141	C536379	D	Discount	-1	2010-12-01 09:41:00	27.50	14527.00	United Kingdom
154	C536383	35004C	SET OF 3 COLOURED FLYING DUCKS	-1	2010-12-01 09:49:00	4.65	15311.00	United Kingdom
235	C536391	22556	PLASTERS IN TIN CIRCUS PARADE	-12	2010-12-01 10:24:00	1.65	17548.00	United Kingdom
236	C536391	21984	PACK OF 12 PINK PAISLEY TISSUES	-24	2010-12-01 10:24:00	0.29	17548.00	United Kingdom
237	C536391	21983	PACK OF 12 BLUE PAISLEY TISSUES	-24	2010-12-01 10:24:00	0.29	17548.00	United Kingdom

```
StockCode
POST      1100
15056BL   291
M          290
C2         133
DOT        16
BANK CHARGES 12
PADS       4
Name: count, dtype: int64
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
112	536381	71270	PHOTO CLIP LINE	1	2010-12-01	1.25	15311	United Kingdom
124	536381	71270	PHOTO CLIP LINE	3	2010-12-01	1.25	15311	United Kingdom
471	536409	90199C	5 STRAND GLASS NECKLACE CRYSTAL	3	2010-12-01	6.35	17908	United Kingdom
486	536409	90199C	5 STRAND GLASS NECKLACE CRYSTAL	1	2010-12-01	6.35	17908	United Kingdom
512	536409	90199C	5 STRAND GLASS NECKLACE CRYSTAL	2	2010-12-01	6.35	17908	United Kingdom
479	536409	85116	BLACK CANDELABRA T-LIGHT HOLDER	1	2010-12-01	2.10	17908	United Kingdom
490	536409	85116	BLACK CANDELABRA T-LIGHT HOLDER	5	2010-12-01	2.10	17908	United Kingdom
548	536412	21448	12 DAISY PEGS IN WOOD BOX	2	2010-12-01	1.65	17920	United Kingdom
561	536412	21448	12 DAISY PEGS IN WOOD BOX	1	2010-12-01	1.65	17920	United Kingdom
556	536412	22243	5 HOOK HANGER RED MAGIC TOADSTOOL	6	2010-12-01	1.65	17920	United Kingdom
559	536412	22243	5 HOOK HANGER RED MAGIC TOADSTOOL	3	2010-12-01	1.65	17920	United Kingdom
566	536412	22243	5 HOOK HANGER RED MAGIC TOADSTOOL	1	2010-12-01	1.65	17920	United Kingdom
545	536412	22077	6 RIBBONS RUSTIC CHARM	1	2010-12-01	1.65	17920	United Kingdom
553	536412	22077	6 RIBBONS RUSTIC CHARM	2	2010-12-01	1.65	17920	United Kingdom
560	536412	22077	6 RIBBONS RUSTIC CHARM	7	2010-12-01	1.65	17920	United Kingdom
558	536412	22144	CHRISTMAS CRAFT LITTLE FRIENDS	2	2010-12-01	2.10	17920	United Kingdom
575	536412	22144	CHRISTMAS CRAFT LITTLE FRIENDS	1	2010-12-01	2.10	17920	United Kingdom
544	536412	21738	COSY SLIPPER SHOES SMALL RED	1	2010-12-01	2.95	17920	United Kingdom
554	536412	21738	COSY SLIPPER SHOES SMALL RED	2	2010-12-01	2.95	17920	United Kingdom
568	536412	21738	COSY SLIPPER SHOES SMALL RED	5	2010-12-01	2.95	17920	United Kingdom

```

# Cheking difference between Description and StockCode
print('Description unique values:', df['Description'].nunique())
print('StockCode unique values:', df['StockCode'].nunique())

✓ 0.0s

Description unique values: 3846
StockCode unique values: 3655

```

Figure A.1 - UnitPrice and Quantity boxplot before outlier removal.

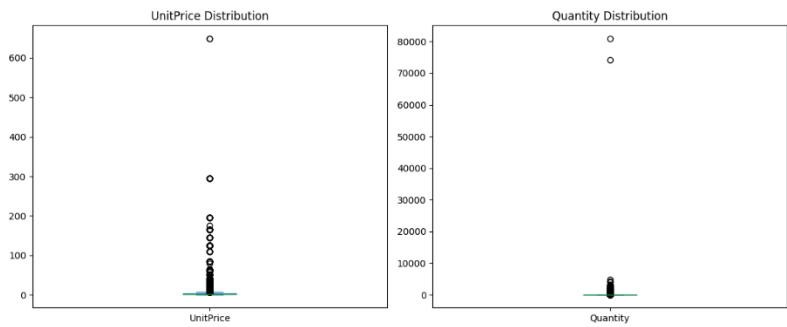


Figure A.2 - UnitPrice and Quantity boxplot after outlier removal.

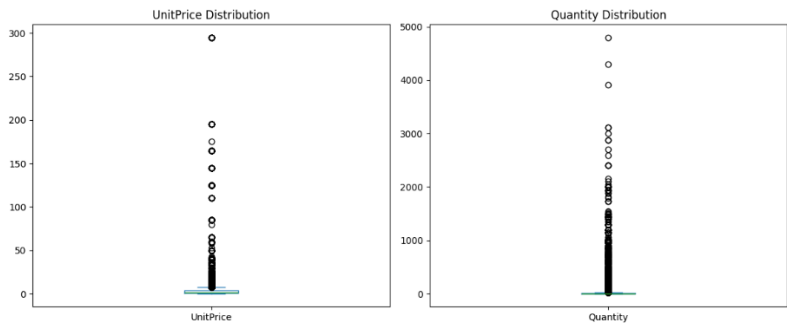


Figure A.3 - Total number of orders and revenue by month.

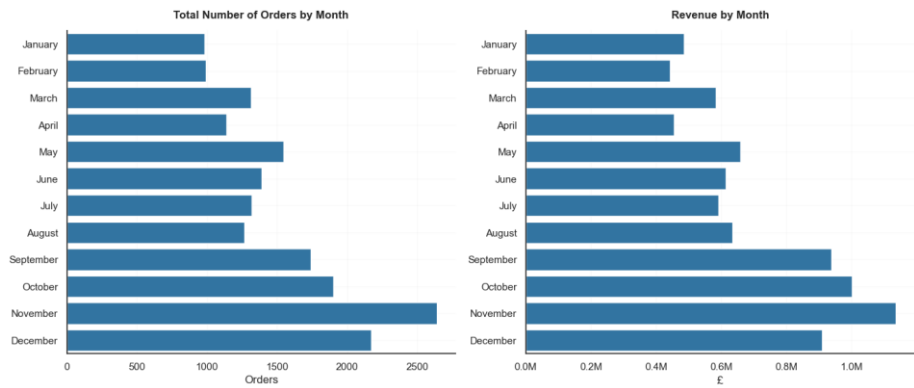


Figure A.4 - Top 10 products purchased by quantity and by revenue.

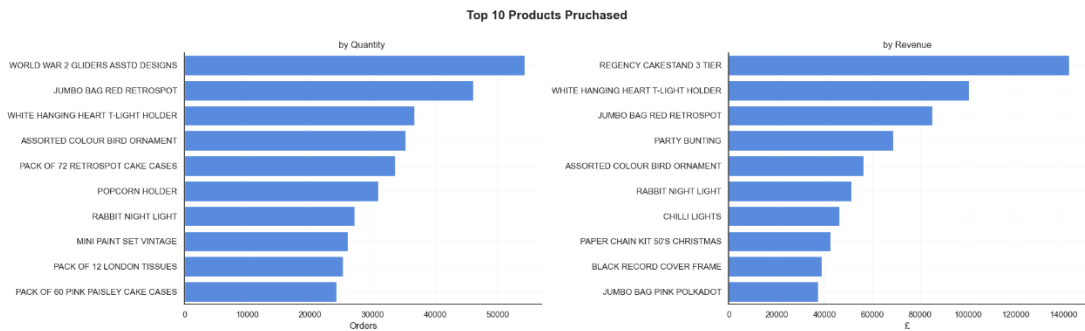


Figure A.5 - Top 30 customers by total orders and revenue.

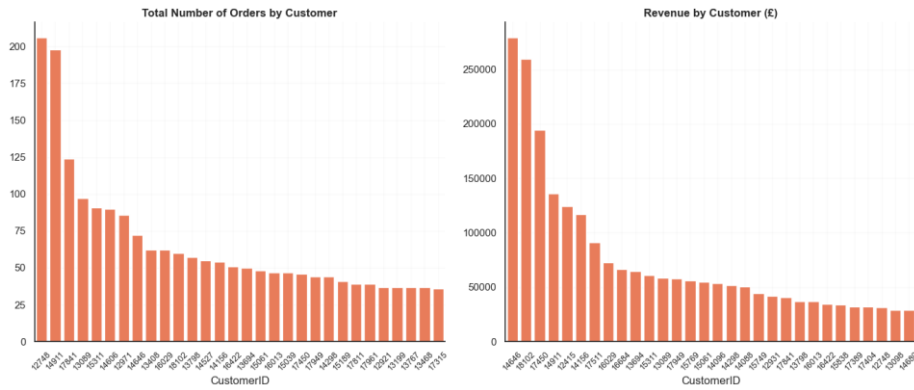


Figure A.6 - Total orders and revenue by country.

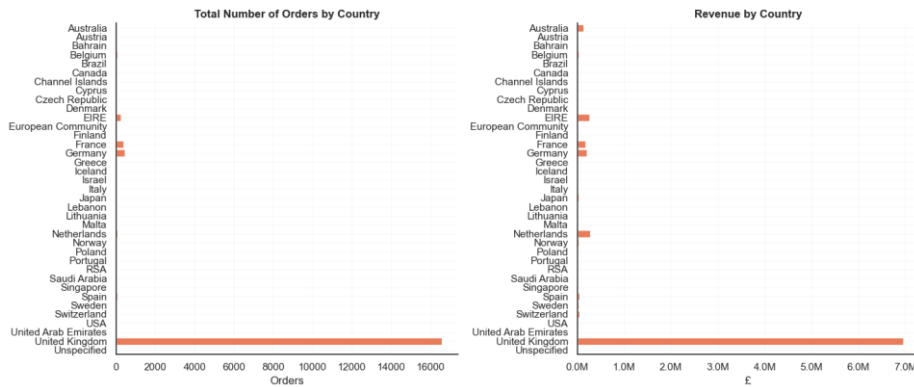


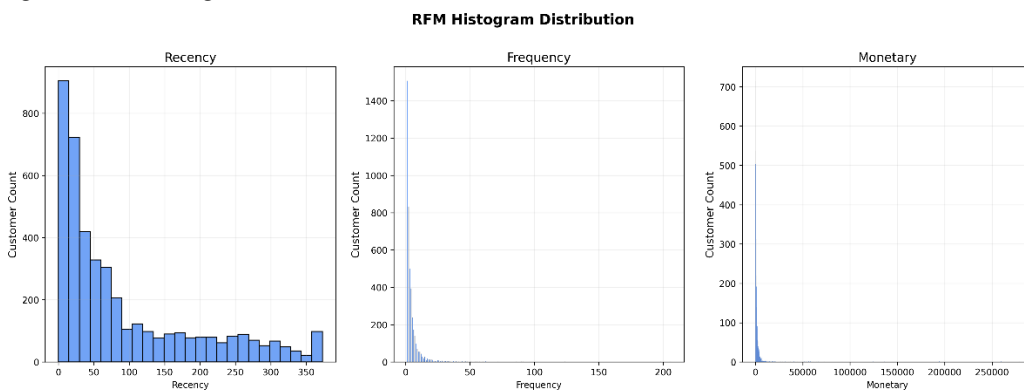
Table A.5 - RFM feature engineering code.

```
# Monetary and Frequency + Last Invoice by customerId
rfm_df = data.groupby(by='CustomerId', as_index=False) \
    .agg(
        Monetary=("TotalPrice", "sum"),
        Frequency=("InvoiceNo", "nunique"),
        LastInvoice=("InvoiceDate", "max")
    )

# Recency will be created using the last invoice date of the year as the current date
max_invoice_date = rfm_df['LastInvoice'].max()

# Recency is the calculation of the last order a customer did comparing the "current" date
rfm_df['Recency'] = (max_invoice_date - rfm_df['LastInvoice']).dt.days
```

Figure A.7 - RFM histogram distribution.



B. Appendix B – Customer Segmentation

Figure B.1 - RFM mean distribution for C1 baseline configuration, $k=3$.

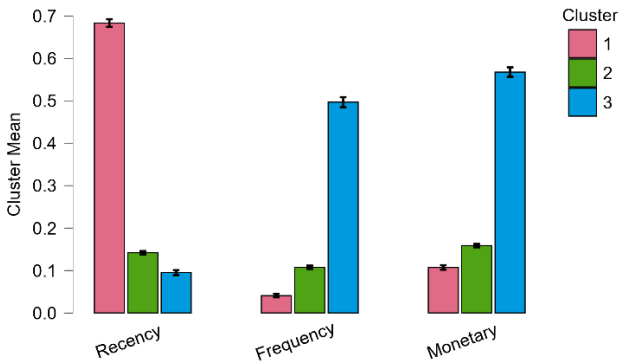


Figure B.2 - RFM mean distribution for C5.2 configuration - $k=4$.

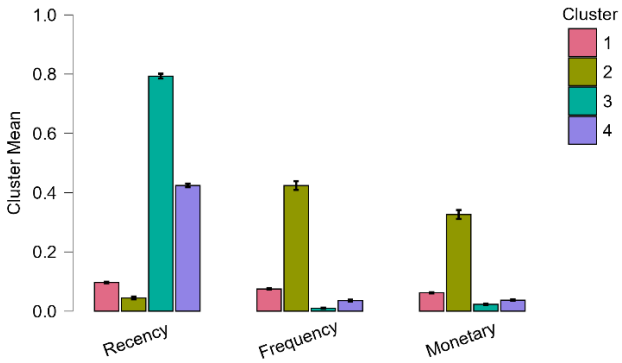


Figure B.3 - Three-dimensional RFM scatterplot - C1 baseline configuration.

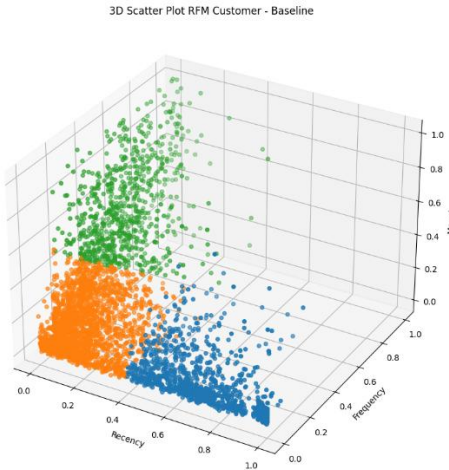


Figure B.4 - Three-dimensional RFM scatterplot - C5.1 (Chosen configuration).

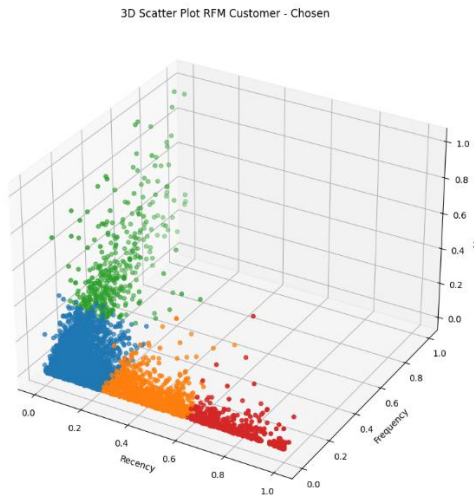


Figure B.5 - Three-dimensional RFM scatterplot - C5.2.

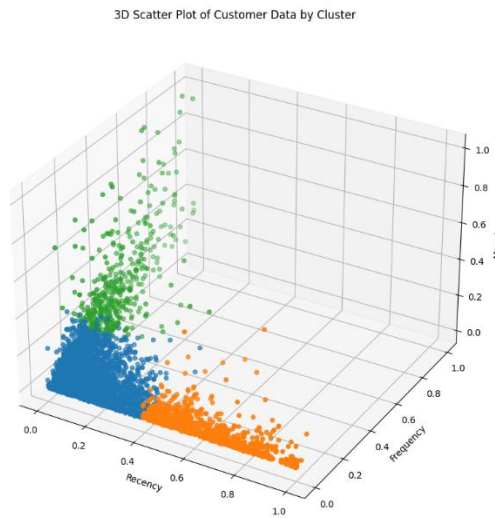


Table B.1 – Cluster RFM means, medians and distribution – C5.1 configuration.

CLUSTER RFM MEANS			
Cluster	Recency	Frequency	Monetary
0	36.11	3.18	1074.41
1	158.25	2.04	646.75
2	16.67	13.30	5640.39
3	295.80	1.29	404.04
CLUSTER RFM MEDIANS			
Cluster	Recency	Frequency	Monetary
0	30.00	3.00	760.75
1	157.00	2.00	416.54
2	10.00	12.00	4762.32
3	288.00	1.00	280.55
CLUSTER DISTRIBUTION (%)			
Cluster			
0	58.31		
1	18.03		
2	9.39		
3	14.26		

Figure B.6 - Three-dimensional RFM highlighting reintegrated outlier customers.

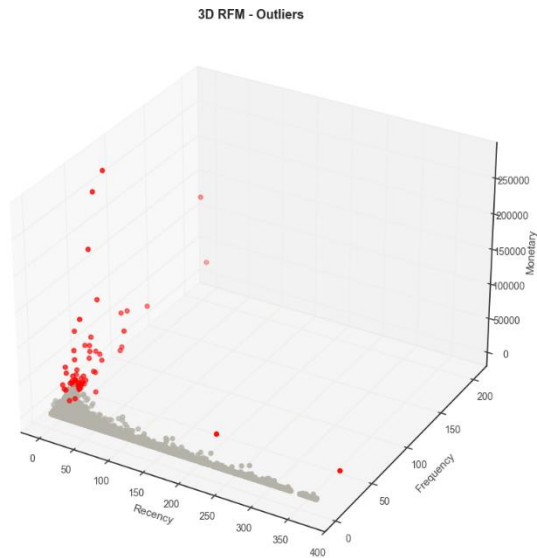


Table B.2 - Cluster RFM means, medians and distribution after manual refinement.

CLUSTER RFM MEANS			
	Recency	Frequency	Monetary
Cluster			
0.00	32.76	3.96	1318.44
1.00	158.25	2.04	646.75
2.00	16.67	13.30	5640.39
3.00	295.80	1.29	404.04
4.00	6.26	45.07	44551.23
5.00	45.43	1.00	396.40
CLUSTER RFM MEDIANS			
	Recency	Frequency	Monetary
Cluster			
0.00	25.00	4.00	1043.41
1.00	157.00	2.00	416.54
2.00	10.00	12.00	4762.32
3.00	288.00	1.00	280.55
4.00	3.00	34.00	28117.04
5.00	46.00	1.00	290.66
CLUSTER DISTRIBUTION (%)			
Cluster			
0.00	42.28		
1.00	17.78		
2.00	9.26		
3.00	14.06		
4.00	1.41		
5.00	15.22		

Table B.3 - Descriptive statistic for RFM features by segment.

	Recency						Frequency						Monetary					
	count	mean	median	min	max	std	count	mean	median	min	max	std	count	mean	median	min	max	std
Cluster_Label																		
Cluster 0	1831	32.76	25.00	0.00	108.00	25.87	1831	3.96	4.00	2.00	10.00	1.91	1831	1318.44	1043.41	36.56	7829.89	983.54
Cluster 1	770	158.25	157.00	94.00	227.00	37.56	770	2.04	2.00	1.00	12.00	1.41	770	646.75	416.54	2.90	9341.26	724.03
Cluster 2	401	16.67	10.00	0.00	163.00	21.13	401	13.30	12.00	3.00	30.00	5.18	401	5640.39	4762.32	1535.77	17286.86	3165.79
Cluster 3	609	295.80	288.00	227.00	373.00	44.17	609	1.29	1.00	1.00	8.00	0.78	609	404.04	280.55	3.75	7832.47	560.74
Cluster 4	63	15.70	3.00	0.00	372.00	54.50	63	44.22	34.00	3.00	206.00	36.72	63	43929.37	28117.04	1296.44	279138.02	53956.43
Cluster 5	659	45.43	46.00	0.00	93.00	24.66	659	1.00	1.00	1.00	1.00	0.00	659	396.40	290.66	5.90	6207.67	506.12

C. Appendix C – Market Basket Analysis

Table C.1 - Top 10 rules by lift - full dataset.

Full Dataset - Top 10 Rules by Lift					
antecedents	consequents	support	confidence	lift	
REGENCY TEA PLATE ROSES, REGENCY TEA PLATE GREEN	REGENCY TEA PLATE PINK	0.0101	0.8079	66.2085	
REGENCY TEA PLATE PINK	REGENCY TEA PLATE ROSES, REGENCY TEA PLATE GREEN	0.0101	0.8259	66.2085	
REGENCY TEA PLATE PINK, REGENCY TEA PLATE ROSES	REGENCY TEA PLATE GREEN	0.0101	0.9391	63.8509	
REGENCY TEA PLATE GREEN	REGENCY TEA PLATE PINK, REGENCY TEA PLATE ROSES	0.0101	0.6852	63.8509	
REGENCY TEA PLATE GREEN	REGENCY TEA PLATE PINK	0.0110	0.7481	61.3147	
REGENCY TEA PLATE PINK	REGENCY TEA PLATE GREEN	0.0110	0.9018	61.3147	
POPPY'S PLAYHOUSE LIVINGROOM	POPPY'S PLAYHOUSE BEDROOM, POPPY'S PLAYHOUSE KITCHEN	0.0101	0.7381	53.3463	
POPPY'S PLAYHOUSE BEDROOM, POPPY'S PLAYHOUSE KITCHEN	POPPY'S PLAYHOUSE LIVINGROOM	0.0101	0.7323	53.3463	
REGENCY MILK JUG PINK	REGENCY SUGAR BOWL GREEN	0.0112	0.7574	51.8787	
REGENCY SUGAR BOWL GREEN	REGENCY MILK JUG PINK	0.0112	0.7687	51.8787	

Table C.2 - Top 10 rules by lift - champions.

Champions - Top 10 Rules by Lift					
antecedents	consequents	support	confidence	lift	
REGENCY MILK JUG PINK	REGENCY SUGAR BOWL GREEN	0.0113	0.7595	58.7012	
REGENCY SUGAR BOWL GREEN	REGENCY MILK JUG PINK	0.0113	0.8696	58.7012	
SET/10 PINK POLKADOT PARTY CANDLES	SET/10 BLUE POLKADOT PARTY CANDLES	0.0103	0.6707	53.3882	
SET/10 BLUE POLKADOT PARTY CANDLES	SET/10 PINK POLKADOT PARTY CANDLES	0.0103	0.8209	53.3882	
REGENCY TEA PLATE ROSES	REGENCY TEA PLATE GREEN	0.0113	0.6316	51.0335	
REGENCY TEA PLATE GREEN	REGENCY TEA PLATE ROSES	0.0113	0.9091	51.0335	
JAM JAR WITH PINK LID	JAM JAR WITH GREEN LID	0.0109	0.7342	50.1970	
JAM JAR WITH GREEN LID	JAM JAR WITH PINK LID	0.0109	0.7436	50.1970	
REGENCY TEA PLATE PINK	REGENCY TEA PLATE ROSES	0.0103	0.8871	49.7988	
POPPY'S PLAYHOUSE LIVINGROOM	POPPY'S PLAYHOUSE BEDROOM, POPPY'S PLAYHOUSE KITCHEN	0.0122	0.7471	48.0051	

Table C.3 - Top 10 rules by lift - inactive.

Inactive - Top 10 Rules by Lift					
antecedents	consequents	support	confidence	lift	
HERB MARKER MINT, HERB MARKER THYME	HERB MARKER ROSEMARY	0.0102	1.0000	98.1250	
HERB MARKER MINT, HERB MARKER THYME	HERB MARKER PARSLEY, HERB MARKER ROSEMARY	0.0102	1.0000	98.1250	
HERB MARKER ROSEMARY	HERB MARKER PARSLEY, HERB MARKER THYME	0.0102	1.0000	98.1250	
HERB MARKER PARSLEY, HERB MARKER ROSEMARY	HERB MARKER MINT, HERB MARKER THYME	0.0102	1.0000	98.1250	
HERB MARKER PARSLEY, HERB MARKER THYME	HERB MARKER MINT, HERB MARKER ROSEMARY	0.0102	1.0000	98.1250	
HERB MARKER MINT, HERB MARKER ROSEMARY	HERB MARKER PARSLEY, HERB MARKER THYME	0.0102	1.0000	98.1250	
HERB MARKER PARSLEY, HERB MARKER THYME	HERB MARKER ROSEMARY	0.0102	1.0000	98.1250	
HERB MARKER PARSLEY, HERB MARKER MINT, HERB MARKER THYME	HERB MARKER ROSEMARY	0.0102	1.0000	98.1250	
HERB MARKER ROSEMARY	HERB MARKER MINT, HERB MARKER THYME	0.0102	1.0000	98.1250	
HERB MARKER ROSEMARY	HERB MARKER PARSLEY, HERB MARKER MINT, HERB MARKER THYME	0.0102	1.0000	98.1250	

Table C.4 - Top 10 rules by lift - lost.

Lost - Top 10 Rules by Lift					
antecedents	consequents	support	confidence	lift	
HERB MARKER MINT, HERB MARKER ROSEMARY	HERB MARKER CHIVES, HERB MARKER THYME	0.0102	1.0000	98.5000	
DECORATION PINK CHICK MAGIC GARDEN	DECORATION BUTTERFLY MAGIC GARDEN	0.0102	1.0000	98.5000	
HERB MARKER MINT	HERB MARKER CHIVES, HERB MARKER ROSEMARY, HERB MARKER THYME	0.0102	1.0000	98.5000	
HERB MARKER CHIVES, HERB MARKER ROSEMARY, HERB MARKER THYME	HERB MARKER MINT	0.0102	1.0000	98.5000	
HERB MARKER MINT	HERB MARKER CHIVES, HERB MARKER THYME	0.0102	1.0000	98.5000	
HERB MARKER CHIVES, HERB MARKER THYME	HERB MARKER MINT	0.0102	1.0000	98.5000	
HERB MARKER CHIVES, HERB MARKER THYME	HERB MARKER MINT, HERB MARKER ROSEMARY	0.0102	1.0000	98.5000	
HERB MARKER ROSEMARY, HERB MARKER CHIVES	HERB MARKER MINT, HERB MARKER THYME	0.0102	1.0000	98.5000	
DECORATION BUTTERFLY MAGIC GARDEN	DECORATION PINK CHICK MAGIC GARDEN	0.0102	1.0000	98.5000	
HERB MARKER MINT	HERB MARKER ROSEMARY, HERB MARKER CHIVES	0.0102	1.0000	98.5000	

Table C.5 - Top 10 rules by lift - new.

New - Top 10 Rules by Lift					
antecedents		consequents	support	confidence	lift
REGENCY TEA PLATE PINK, GREEN REGENCY TEACUP AND SAUCER		PINK REGENCY TEACUP AND SAUCER, REGENCY TEA PLATE GREEN	0.0121	1.0000	82.3750
PINK REGENCY TEACUP AND SAUCER, REGENCY TEA PLATE GREEN		REGENCY TEA PLATE PINK, GREEN REGENCY TEACUP AND SAUCER	0.0121	1.0000	82.3750
PINK REGENCY TEACUP AND SAUCER, REGENCY TEA PLATE GREEN		REGENCY TEA PLATE PINK	0.0121	1.0000	73.2222
REGENCY TEA PLATE PINK		PINK REGENCY TEACUP AND SAUCER, REGENCY TEA PLATE GREEN	0.0121	0.8889	73.2222
REGENCY TEA PLATE PINK		GREEN REGENCY TEACUP AND SAUCER, REGENCY TEA PLATE GREEN, PINK REGENCY TEACUP AND SAUCER	0.0121	0.8889	73.2222
GREEN REGENCY TEACUP AND SAUCER, REGENCY TEA PLATE GREEN, PINK REGENCY TEACUP AND SAUCER		REGENCY TEA PLATE PINK	0.0121	1.0000	73.2222
SMALL RED RETROSPOT MUG IN BOX		SMALL WHITE RETROSPOT MUG IN BOX	0.0106	0.7778	73.2222
SMALL WHITE RETROSPOT MUG IN BOX		SMALL RED RETROSPOT MUG IN BOX	0.0106	1.0000	73.2222
REGENCY TEA PLATE PINK, PINK REGENCY TEACUP AND SAUCER		GREEN REGENCY TEACUP AND SAUCER, REGENCY TEA PLATE GREEN	0.0121	1.0000	73.2222
GREEN REGENCY TEACUP AND SAUCER, REGENCY TEA PLATE GREEN		REGENCY TEA PLATE PINK, PINK REGENCY TEACUP AND SAUCER	0.0121	0.8889	73.2222

Table C.6 - Top 10 rules by lift - occasional.

Occasional - Top 10 Rules by Lift					
antecedents		consequents	support	confidence	lift
HERB MARKER THYME		HERB MARKER PARSLEY	0.0103	0.9615	88.3520
HERB MARKER PARSLEY		HERB MARKER THYME	0.0103	0.9494	88.3520
SET/6 RED SPOTTY PAPER CLIPS	SET/6 RED SPOTTY PAPER PLATES		0.0113	0.8913	63.4312
SET/6 RED SPOTTY PAPER PLATES	SET/6 RED SPOTTY PAPER CUPS		0.0113	0.8039	63.4312
REGENCY SUGAR BOWL GREEN		REGENCY MILK JUG PINK	0.0103	0.8242	59.2346
REGENCY MILK JUG PINK		REGENCY SUGAR BOWL GREEN	0.0103	0.7426	59.2346
REGENCY TEA PLATE GREEN		REGENCY TEA PLATE ROSES	0.0103	0.8242	56.4405
REGENCY TEA PLATE ROSES		REGENCY TEA PLATE GREEN	0.0103	0.7075	56.4405
COFFEE MUG PEARS DESIGN		COFFEE MUG APPLES DESIGN	0.0103	0.8065	53.7067
COFFEE MUG APPLES DESIGN		COFFEE MUG PEARS DESIGN	0.0103	0.6881	53.7067

Table C.7 - Top 10 rules by lift - VIP.

VIP - Top 10 Rules by Lift					
antecedents		consequents	support	confidence	lift
HERB MARKER ROSEMARY		HERB MARKER PARSLEY	0.0105	0.9667	91.6333
HERB MARKER PARSLEY		HERB MARKER ROSEMARY	0.0105	1.0000	91.6333
HERB MARKER ROSEMARY		HERB MARKER BASIL	0.0102	0.9333	85.5244
HERB MARKER BASIL		HERB MARKER ROSEMARY	0.0102	0.9333	85.5244
SET OF 3 WOODEN HEART DECORATIONS, SET OF 3 WOODEN SLEIGH DECORATIONS		SET OF 3 WOODEN TREE DECORATIONS, SET OF 3 WOODEN STOCKING DECORATION	0.0135	0.8810	57.6604
SET OF 3 WOODEN TREE DECORATIONS, SET OF 3 WOODEN STOCKING DECORATION		SET OF 3 WOODEN HEART DECORATIONS, SET OF 3 WOODEN SLEIGH DECORATIONS	0.0135	0.8810	57.6604
SET OF 3 WOODEN TREE DECORATIONS, SET OF 3 WOODEN SLEIGH DECORATIONS		SET OF 3 WOODEN HEART DECORATIONS, SET OF 3 WOODEN STOCKING DECORATION	0.0135	0.9024	56.3819
SET OF 3 WOODEN HEART DECORATIONS, SET OF 3 WOODEN STOCKING DECORATION		SET OF 3 WOODEN TREE DECORATIONS, SET OF 3 WOODEN SLEIGH DECORATIONS	0.0135	0.8409	56.3819
REGENCY TEA PLATE GREEN, ROSES REGENCY TEACUP AND SAUCER		GREEN REGENCY TEACUP AND SAUCER, REGENCY TEA PLATE ROSES	0.0109	0.7143	53.0695
GREEN REGENCY TEACUP AND SAUCER, REGENCY TEA PLATE ROSES		REGENCY TEA PLATE GREEN, ROSES REGENCY TEACUP AND SAUCER	0.0109	0.8108	53.0695